# Survival Analysis

Lecture notes, Lectures 1–10

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# Lecture 1. Introduction

## 1 Course Overview

- Textbook: Klein & Moeschberger, *Survival Analysis: Techniques for Censored and Truncated Data*, 2nd ed. (cited below as K&M). These notes follow its Chapters 2–5, 7–9, 11 and 12.
- The emphasis is on methodology and application, with part of the underlying theory (why the methods work and how they were developed).
- Prerequisites: basic probability theory, statistical inference, and R programming.

## 2 What Is Survival Data?

**Definition 1.1: Survival data**

**Survival data** track the *time until an event of interest occurs*:

$$\text{Exposure (clock starts)} \xrightarrow{\text{survival time}} \text{Event (clock ends)}.$$

Examples: time from diagnosis to death; time from treatment to cancer recurrence; time to failure of a machine or component; time to staff promotion; time to divorce; time to unsubscription.

**Example 1.2: Worcester Heart Attack Study (WHAS100)**

Survival times following hospitalisation for a heart attack (MI). A snapshot:

| ID | Female | Age (years) | Follow-up (years) | Event (death) |
|---|---|---|---|---|
| 16 | 0 | 39 | 7.2 | 0 |
| 66 | 1 | 91 | 1.0 | 1 |
| 71 | 0 | 65 | 5.1 | 0 |
| 55 | 0 | 80 | 5.3 | 0 |
| 61 | 0 | 90 | 0.3 | 1 |

**Question:** what is the mean survival time after hospitalisation?

- *Wrong approach 1* (use only subjects with known survival times): $(1.0 + 0.3)/2 = 0.65$. This is an **underestimate**, because we ignored the people who survived longer.
- *Wrong approach 2* (pretend follow-up time = survival time): $(7.2 + 1.0 + 5.1 + 5.3 + 0.3)/5 = 3.78$. Also an **underestimate**, because IDs 16, 71, 55 actually survived *longer* than their follow-up times.

**Acknowledging censoring.** Writing censored times with a "+": $7.2^+$, 1.0, $5.1^+$, $5.3^+$, 0.3. For ID 66 we know the patient survived exactly 1 year after MI; for ID 16 we only know that the patient survived *at least* 7.2 years. The data carry **incomplete (partial) information**.

**Remark 1.3: Why both naive estimates are biased downward**

Let $T_i$ be the true survival time and $C_i$ the follow-up (censoring) time, so we observe $Y_i = \min(T_i, C_i)$. Since $Y_i \le T_i$ always, $\mathrm{E}(\bar{Y}) \le \mathrm{E}(T)$ (approach 2). Approach 1 keeps only $\{T_i \le C_i\}$, i.e. it conditions on the event time being *small enough* to be seen, which preferentially

discards long survivors.

## 3 What Is Special About Survival Data?

- Classical mathematical statistics is built on i.i.d. data, e.g. $n$ i.i.d. copies $T_1, \ldots, T_n$ of the survival time.
- The **incompleteness** of survival data distinguishes survival analysis from traditional methods.
- Formulating a survival problem requires three ingredients:
  (1) the **event of interest**;
  (2) the **time origin**;
  (3) the **time scale**.
- Incompleteness is realized as either **censoring** or **truncation**. Common types: right censoring (RC), left censoring (LC), interval censoring (IC), left truncation (LT), right truncation (RT).

## 4 Types of Censoring

### 4.1 Right censoring

**Definition 1.4: Right censoring (RC)**

For a right-censored observation, the exact event time is unknown; all we know is that the event happens *after* a certain time point: $T_i > C_i$.

**Example 1.5: Time to death in a breast cancer trial**

$T_i$ = time to death from randomization (receipt of treatment); the time of randomization is the time origin. If the study ends 9 months after the patient's treatment date and the patient is still alive, $T_i$ is right censored at 9 months: $T_i > 9$.

### 4.2 Left censoring

**Definition 1.6: Left censoring (LC)**

For a left-censored observation, all we know is that the event happened *prior to* a certain time point: $T_i < C_i$.

**Example 1.7: Time to first use of marijuana**

Question: "When did you first use marijuana?" Possible answers:

- the exact age (uncensored);
- "never used it" (right censored at the current age);
- "used it but cannot recall when" (left censored at the current age).

When both left and right censoring are present in the data we speak of **double censoring**.

### 4.3 Interval censoring

**Definition 1.8: Interval censoring (IC)**

For an interval-censored observation, all we know is that the event happened within an interval: $T_i \in (L_i, R_i]$.

**Example 1.9: Time to tumour recurrence**

After breast cancer surgery, MRI is performed every 5 months. If the tumour is absent at month 10 and detected at month 15, then $10 < T_i \leq 15$.

**Remark 1.10**

RC and LC are special cases of IC: right censoring at $C$ is $T \in (C, \infty)$, left censoring at $C$ is $T \in [0, C)$.

## 5 Types of Truncation

### 5.1 Left truncation

**Definition 1.11: Left truncation (LT)**

Only individuals who "survive" a sufficient time are **included** in the sample. If $V_i$ is the entry (truncation) time, $T_i$ is observed only if $T_i \geq V_i$, i.e. we observe the conditional distribution of $T_i$ given $T_i \geq V_i$.

**Example 1.12: Longevity of elderly residents of a retirement community**

An individual must survive to a sufficient age to enter the retirement center; anyone who died earlier never enters. The data are left truncated. Ignoring LT leads to **length-biased sampling**: long lifetimes are over-represented.

### 5.2 Right truncation

**Definition 1.13: Right truncation (RT)**

Only individuals who have *experienced the event* by a specified time are included in the sample: $T_i$ is observed only if $T_i < R_i$.

**Example 1.14: Age at onset of diabetes**

Diabetic patients are enrolled and asked retrospectively for their age of onset. With $R_i =$ age at the time of the study and $T_i =$ age of onset, $T_i$ is observed conditionally on $T_i < R_i$: a subject must already have diabetes to be included.

**Example 1.15: HIV incubation time (Lagakos et al., 1988)**

Data on 258 adults and 37 children infected with HIV who developed AIDS by June 30, 1986. With $X$ = age at HIV infection and $R$ = age at developing AIDS, the incubation time $R - X$ is known only for those who were infected *and* developed AIDS within the study window. This is right truncation.

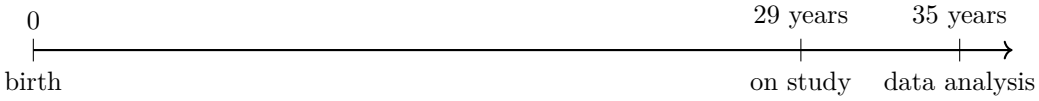
### 5.3 Censoring vs. truncation

| | |
|---|---|
| **Censoring** | The event/subject *is detected*, but its value is not known completely: we only know it is less than and/or more than some number. |
| **Truncation** | A subject is detected *only if* its value is greater (or less) than some number; when detected, its value is completely known. |

The key difference: a censored object is detectable, whereas a truncated object is not even detectable (it never enters the data set).

## 6 CVD Example: Time Scales, Truncation and Censoring Together

An observational study enrols subjects *without* heart disease to identify risk factors for cardiovascular disease (CVD); the event is CVD. Consider a subject who enters at age 29; data are analyzed when the subject would be 35.




Two time scales are possible depending on the origin:

- origin = birth: $T_i^1$ = age at CVD;
- origin = study entry: $T_i^2$ = years on study until CVD.

Both may be right censored (subject still CVD-free at the end of study). The first scale $T_i^1$ is usually more interesting.

| Scenario | $T_i^1$ (age) | $T_i^2$ (time on study) |
|---|---|---|
| CVD at age 34 | $T_i^1 = 34$ observed *given* $T_i^1 \geq 29$: **LT at 29** | $T_i^2 = 34 - 29 = 5$: no truncation, no censoring |
| No CVD by 35 | $T_i^1 > 35$ given $T_i^1 \geq 29$: **LT at 29 and RC at 35** | $T_i^2 > 6$: **RC**, not LT |
| Enrolled at 30, already had CVD before 30 | $T_i^1 < 30$: **LC at 30** | — |

## 7 Definitions and Censoring Patterns

- **Failure time** (survival time, event time): the length of "time" to an event.
- **Time origin**: objectively defined and scientifically meaningful; ideally all subjects are comparable at the origin.
- **Time scale**: usually clock/calendar time (seconds, hours, weeks, months, years), but others are possible, e.g. QALY (quality-adjusted life years).

**Definition 1.16: Equal or unequal censoring times**

- **Type I censoring**: all subjects enter at the same real time and all censoring times are equal.
- **Generalized Type I censoring**: with staggered entry the censoring times differ, but each is known at entry.
- **Random censoring**: losses to follow-up (death from other causes, relocation, withdrawal of consent); censoring times are known only for those actually censored, and they differ.

It is crucial that the actual censoring *times* are recorded, not just the fact that a subject is censored.

## 8 Possible Simple (Naive) Methods

1. **Dichotomise at a pre-specified $t_0$.** Survivors to $t_0$ are "successes", deaths before $t_0$ "failures"; use binary-data methods (chi-square test, logistic regression).
   Disadvantages: difficulty with censoring before $t_0$; loss of efficiency from concentrating on a single point of the survival curve; arbitrary choice of $t_0$.
2. **Take logs and use the linear model.** Disadvantages: possible undue dependence on short failure times; the log-normal distribution may not be appropriate; still cannot handle censoring.

Both can be useful as a prelude to more complex analyses.

## 9 Summary

- Six types of incompleteness: RC, LC, IC, LT, RT (plus double censoring). LT and RC are the most common sampling schemes.
- **Censoring only**: $T_i$ may not be observed directly; it is known to be $>$ or $<$ some value, or to lie in an interval.
- **Truncation only**: $T_i$ is observed conditionally on $T_i >$ or $<$ some value.
- Mixtures of censoring and truncation are more complex.

# Lecture 2. Basic Quantities

## Contents

## 1 Review: Classifying Incomplete Data

| Situation | Type |
|---|---|
| Clinical trial starts at randomization (time 0); a participant completes 24 months with no event by study end. | RC |
| At baseline the tumour is already present; exact onset unknown, only that it occurred before baseline. | LC |
| Annual screening; disease-free at year 2, first positive at year 3. | IC |
| Registry enrols patients only if alive at clinic entry (some enter 3 years after diagnosis); followed until death or end of study. | LT + (RC or event) |
| Historical cohort assembled in 2020 from obituaries of people who died in 2010–2019; age at death recorded, no survivors. | RT |

## 2 Survival Function

Let $X \geq 0$ be a random failure time.

**Definition 2.1: Survival function**

For continuous $X$ with pdf $f$ and cdf $F(x) = P(X \leq x)$,

$$S(x) = P(X > x) = 1 - F(x) = \int_x^\infty f(s)\,\mathrm{d}s.$$

$S(x)$ is the probability of surviving beyond time $x$. It is monotone non-increasing with $S(0) = 1$ and $S(\infty) = 0$.

Graphically, $S(x)$ is the area under the density to the right of $x$. (Typical examples are the Exp(0.5) density and the 1989 US all-cause survival curves by race/sex, which stay near 1 until middle age and then drop steeply.)

## 3 Hazard Function

**Definition 2.2: Hazard function**

$$h(x) = \lim_{\Delta x \to 0} \frac{P(x \leq X < x + \Delta x \mid X \geq x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{P(x \leq X < x + \Delta x)/P(X \geq x)}{\Delta x} = \frac{f(x)}{S(x)}.$$

- $f(x)\,\mathrm{d}x \approx P(x \leq X < x + \mathrm{d}x) = S(x) - S(x + \mathrm{d}x)$ is the "infinitesimal" probability of failure at $x$, *unconditionally* on being alive before $x$.
- $h(x)\,\mathrm{d}x$ is the infinitesimal probability of failure in the next instant after $x$, *given* alive at $x$.

**Remark 2.3: $h$ is a rate, not a probability**

$h(x) \geq 0$ can exceed 1; only $h(x)\Delta x$ for small $\Delta x$ approximates a conditional probability.

### 3.1 Shapes of hazard functions

(i) **Constant**: e.g. the exponential distribution.

(ii) **Increasing**: industrial systems and cars, whose component failure rates rise over time (ageing).

(iii) **Decreasing** (less common): e.g. transplant studies (risk is highest just after the operation).

(iv) **Bathtub-shaped**: population mortality (high infant mortality, low in mid-life, rising in old age).

(v) **Hump-shaped (unimodal)**: rises early and eventually declines; e.g. breast cancer treatment studies where surgery- or toxicity-related high-risk patients tend to die early.

### 3.2 Examples

**Example 2.4: Exponential($\lambda$)**

$f(x) = \lambda e^{-\lambda x}$ and

$$S(x) = \int_x^\infty \lambda e^{-\lambda s}\,\mathrm{d}s = \big[-e^{-\lambda s}\big]_x^\infty = e^{-\lambda x}, \qquad h(x) = \frac{f(x)}{S(x)} = \lambda.$$

The constant hazard is the *memoryless* property: $P(X > x + t \mid X > x) = e^{-\lambda t}$ does not depend on $x$.

**Example 2.5: Weibull($\alpha, \lambda$)**

$S(x) = \exp(-\lambda x^\alpha)$. Then

$$f(x) = -\frac{\mathrm{d}S(x)}{\mathrm{d}x} = \alpha\lambda x^{\alpha-1}\exp(-\lambda x^\alpha), \qquad h(x) = \frac{f(x)}{S(x)} = \alpha\lambda x^{\alpha-1}.$$

$\alpha > 1$: increasing hazard; $\alpha = 1$: constant (exponential); $\alpha < 1$: decreasing.

## 4 Cumulative Hazard and the $S$–$h$ Relationship

**Motivation.** Survival, density, hazard and cumulative hazard are four descriptions of the same distribution, and different methods estimate different ones. The relation $S(t) = \exp\{-H(t)\}$ lets us move between them.

**Theorem 2.6: Continuous case**

Define the **cumulative hazard** $H(x) = \int_0^x h(s)\,\mathrm{d}s$. If $X$ is absolutely continuous,

$$H(x) = -\ln S(x), \qquad S(x) = \exp\{-H(x)\} = \exp\Big\{-\int_0^x h(s)\,\mathrm{d}s\Big\}.$$

**Proof [Supplement].** Since $f(s) = -S'(s)$,

$$H(x) = \int_0^x \frac{f(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \frac{S'(s)}{S(s)}\,\mathrm{d}s = -\int_0^x \mathrm{d}\{\ln S(s)\} = -\big[\ln S(x) - \ln S(0)\big] = -\ln S(x),$$

using $S(0) = 1$. Exponentiating gives $S(x) = e^{-H(x)}$. ■

| **Summary of relations (continuous $X$)** | |
|---|---|
| $f(t) = \dfrac{\mathrm{d}F(t)}{\mathrm{d}t} = -S'(t)$ | $F(t) = 1 - S(t)$ |
| $h(t) = \dfrac{f(t)}{S(t)} = -\dfrac{\mathrm{d}}{\mathrm{d}t}\ln S(t)$ | $H(t) = \displaystyle\int_0^t h(s)\,\mathrm{d}s = -\ln S(t)$ |
| $S(t) = \exp\{-H(t)\}$ | $f(t) = h(t)\exp\{-H(t)\}$ |

Any one of $f, F, S, h, H$ determines all the others.

## 5 Discrete Failure Times

At points $0 = x_0 < x_1 < \cdots < x_d$ let $P(X = x_j) = p_j > 0$, $\sum_{j=1}^d p_j = 1$. Discrete times arise from rounding of measurements or grouping failure times into intervals. Then

$$S(x) = P(X > x) = \sum_{j=1}^{d} P(X = x_j)\, I(x_j > x),$$

a right-continuous step function.

**Definition 2.7: Discrete hazard**

$$h(x_j) = P(X = x_j \mid X \geq x_j) = \frac{p_j}{S(x_{j-1})} = \frac{p_j}{\sum_{k=j}^d p_k}, \qquad j = 1, \ldots, d.$$

Here $h(x_j)$ *is* a conditional probability, so $0 < h(x_j) \leq 1$.

Since $p_j = S(x_{j-1}) - S(x_j)$,

$$h(x_j) = \frac{S(x_{j-1}) - S(x_j)}{S(x_{j-1})} = 1 - \frac{S(x_j)}{S(x_{j-1})}.$$

**Example 2.8:** $p_j = 1/3$, $j = 1, 2, 3$

$$S(x) = \begin{cases} 1, & 0 \leq x < 1 \\ 2/3, & 1 \leq x < 2 \\ 1/3, & 2 \leq x < 3 \\ 0, & x \geq 3 \end{cases} \qquad h(1) = \frac{1/3}{1} = \frac{1}{3}, \quad h(2) = \frac{1/3}{2/3} = \frac{1}{2}, \quad h(3) = \frac{1/3}{1/3} = 1.$$

The last hazard equals 1: anyone alive just before the last support point must fail there.

**Theorem 2.9: Product-limit representation**

$$S(x_j) = \frac{S(x_j)}{S(x_{j-1})}\frac{S(x_{j-1})}{S(x_{j-2})}\cdots\frac{S(x_1)}{S(x_0)}S(x_0) = \prod_{i:x_i\le x_j}\frac{S(x_i)}{S(x_{i-1})} = \prod_{i:x_i\le x_j}[1-h(x_i)],$$

using the telescoping product and $S(x_0) = S(0) = 1$.

This product form is the population version of the Kaplan–Meier estimator (Lecture 5).

**Remark 2.10: Two definitions of the discrete cumulative hazard**

Klein & Moeschberger define $H(x_j) = \sum_{i:x_i\le x_j} h(x_i)$. With this definition $S(x) = \exp\{-H(x)\}$ does *not* hold. Cox & Oakes (1984) instead define

$$H(x_j) = -\sum_{i:x_i\le x_j}\log\left[1-h(x_i)\right] \approx \sum_{i:x_i\le x_j} h(x_i),$$

which restores $S = e^{-H}$ exactly (take logs of the product-limit form). The approximation uses $-\log(1-u)\approx u$ for small $u$.

## 6 Mean and Mean Residual Life

**Motivation.** For a patient who has survived to time $t$, the relevant question is the expected remaining lifetime. The mean residual life answers it, and it can be computed from the survival function alone.

**Theorem 2.11: Mean as area under $S$**

For $X\ge 0$, $\mathrm{E}(X) = \int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty S(t)\,\mathrm{d}t$.

**Proof [Supplement].** Write $t = \int_0^t \mathrm{d}u$ and exchange the order of integration (Tonelli, integrand $\ge 0$):

$$\int_0^\infty t f(t)\,\mathrm{d}t = \int_0^\infty\int_0^t f(t)\,\mathrm{d}u\,\mathrm{d}t = \int_0^\infty\int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_0^\infty S(u)\,\mathrm{d}u.$$

■

**Definition 2.12: Mean residual life**

$\mathrm{mrl}(x) = \mathrm{E}(X - x \mid X > x)$. For continuous $X$,

$$\mathrm{mrl}(x) = \frac{\int_x^\infty (t-x)f(t)\,\mathrm{d}t}{S(x)} = \frac{\int_x^\infty S(t)\,\mathrm{d}t}{S(x)},$$

the area under the survival curve to the right of $x$ divided by $S(x)$. Note $\mathrm{mrl}(0) = \mathrm{E}(X)$.

**Proof of the second equality [Supplement].** Same Fubini argument with $t - x = \int_x^t \mathrm{d}u$: $\int_x^\infty (t-x)f(t)\,\mathrm{d}t = \int_x^\infty\int_u^\infty f(t)\,\mathrm{d}t\,\mathrm{d}u = \int_x^\infty S(u)\,\mathrm{d}u$. ■

# 7 Quantiles

**Definition 2.13: $p$-th quantile**

The $p$-th quantile ($100p$-th percentile) $x_p$ satisfies $S(x_p) \le 1-p$:

$$x_p = \inf\{t : S(t) \le 1-p\}.$$

For continuous strictly decreasing $S$ this is the unique solution of $S(x_p) = 1-p$, i.e. $F(x_p) = p$. The median is $x_{0.5}$.

**Example 2.14: Exponential($\lambda$) — mean, mrl, median**

$S(x) = e^{-\lambda x}$.

- $\mathrm{E}(X) = \int_0^\infty e^{-\lambda t}\,\mathrm{d}t = 1/\lambda$.
- $\mathrm{mrl}(x) = \dfrac{\int_x^\infty e^{-\lambda t}\,\mathrm{d}t}{e^{-\lambda x}} = \dfrac{e^{-\lambda x}/\lambda}{e^{-\lambda x}} = \dfrac{1}{\lambda}$ (constant: memorylessness).
- Median: $e^{-\lambda x_{0.5}} = 1/2 \Rightarrow x_{0.5} = (\ln 2)/\lambda$.

**Example 2.15: Weibull quantile**

$S(x) = \exp(-\lambda x^\alpha)$, so $F(x_p) = p$ gives

$$1 - e^{-\lambda x_p^\alpha} = p \implies e^{-\lambda x_p^\alpha} = 1-p \implies \lambda x_p^\alpha = -\ln(1-p) \implies x_p = \Big[\frac{-\ln(1-p)}{\lambda}\Big]^{1/\alpha}.$$

In particular the median is $x_{0.5} = (\ln 2/\lambda)^{1/\alpha}$.

**Example 2.16: LogNormal quantile**

$X \sim \mathrm{LogNormal}(\mu, \sigma^2)$ means $Y = \ln X \sim N(\mu, \sigma^2)$. Since ln is strictly increasing, quantiles map through it:

$$p = P(X \le x_p) = P(Y \le \ln x_p) = \Phi\Big(\frac{\ln x_p - \mu}{\sigma}\Big) \implies x_p = \exp\left(\mu + \sigma\Phi^{-1}(p)\right),$$

where $\Phi^{-1}$ is the standard normal quantile function. With $p = 1/2$, $\Phi^{-1}(1/2) = 0$, so $\mathrm{median}(X) = e^\mu$.

# 8 Median Residual Life

**Definition 2.17: Median residual life**

Preferred to the mrl when the distribution is highly skewed.

$$\operatorname{mdrl}(x) = \operatorname{median}(X - x \mid X > x),$$

equivalently

$$P(X - x > \operatorname{mdrl}(x) \mid X > x) = \frac{1}{2} \iff \frac{P(X > x + \operatorname{mdrl}(x),\, X > x)}{P(X > x)} = \frac{1}{2}.$$

For continuous $X$ the numerator is $S(x + \operatorname{mdrl}(x))$, hence

$$S(x + \operatorname{mdrl}(x)) = \tfrac{1}{2} S(x) \quad \Longrightarrow \quad \operatorname{mdrl}(x) = S^{-1}\Big(\frac{S(x)}{2}\Big) - x.$$

**Example 2.18: Exponential($\lambda$)**

$S^{-1}(y) = -(1/\lambda)\ln y$ for $y \in (0, 1]$, so

$$\operatorname{mdrl}(x) = -\frac{1}{\lambda}\ln\Big(\frac{e^{-\lambda x}}{2}\Big) - x = -\frac{1}{\lambda}\big(-\lambda x - \ln 2\big) - x = \frac{\ln 2}{\lambda}.$$

Again constant in $x$.

## 9 Weibull Mean Residual Life

Let $S(t) = \exp(-\lambda t^{\alpha})$, $\lambda, \alpha > 0$, and $m(t) = \mathrm{E}[T - t \mid T > t]$.

1. Derive $m(t)$. 2. Show $m(t)$ is constant when $\alpha = 1$. 3. For $\alpha > 1$, show $m(t)$ is decreasing and interpret.

**Derivation: substitution $v = \lambda u^{\alpha}$**

With $f(u) = \lambda\alpha u^{\alpha-1} e^{-\lambda u^{\alpha}}$, the numerator is

$$N(t) = \int_t^{\infty} (u - t)\, \lambda\alpha u^{\alpha-1} e^{-\lambda u^{\alpha}}\, \mathrm{d}u.$$

Set $v = \lambda u^{\alpha}$, so $\mathrm{d}v = \lambda\alpha u^{\alpha-1}\,\mathrm{d}u$ and $u = (v/\lambda)^{1/\alpha}$; limits $u = t \mapsto v = \lambda t^{\alpha}$, $u \to \infty \mapsto v \to \infty$. Then

$$\begin{aligned} N(t) &= \int_{\lambda t^{\alpha}}^{\infty} \Big(\big(\tfrac{v}{\lambda}\big)^{1/\alpha} - t\Big) e^{-v}\,\mathrm{d}v = \lambda^{-1/\alpha} \int_{\lambda t^{\alpha}}^{\infty} v^{1/\alpha} e^{-v}\,\mathrm{d}v - t\int_{\lambda t^{\alpha}}^{\infty} e^{-v}\,\mathrm{d}v \\ &= \lambda^{-1/\alpha}\,\Gamma\big(1 + \tfrac{1}{\alpha}, \lambda t^{\alpha}\big) - t\, e^{-\lambda t^{\alpha}}, \end{aligned}$$

where $\Gamma(s, x) = \int_x^{\infty} v^{s-1} e^{-v}\,\mathrm{d}v$ is the upper incomplete gamma function. Dividing by $S(t) = e^{-\lambda t^{\alpha}}$:

$$m(t) = \frac{N(t)}{S(t)} = \frac{1}{S(t)} \cdot \frac{1}{\lambda^{1/\alpha}}\,\Gamma\big(1 + \tfrac{1}{\alpha}, \lambda t^{\alpha}\big) - t.$$

- $\alpha = 1$: $\Gamma(2, \lambda t) = (1 + \lambda t)e^{-\lambda t}$, so $m(t) = \frac{(1+\lambda t)}{\lambda} - t = \frac{1}{\lambda}$ (constant).
- $\alpha > 1$: the hazard $h(t) = \lambda\alpha t^{\alpha-1}$ is increasing, hence $m(t)$ decreases in $t$ (proved in §10). Interpretation: the system *ages* — the longer it has survived, the shorter its expected remaining life.

## 10 mrl and Hazard Monotonicity

**Motivation.** Intuitively, if the risk of failure grows with age, the expected remaining life should shrink. This section makes that precise.

**Theorem 2.19**

Let $T \geq 0$ have survival $S$, density $f$, hazard $h = f/S$, and $m(t) = \mathrm{E}[T - t \mid T > t]$. Then

1. $m'(t) = h(t)m(t) - 1$;
2. if $h$ is increasing then $m$ is decreasing; if $h$ is decreasing then $m$ is increasing.

**(1) Differential equation [Supplement].** $m(t) = \int_t^\infty S(u)\,\mathrm{d}u / S(t)$. By the quotient rule and $\frac{\mathrm{d}}{\mathrm{d}t}\int_t^\infty S(u)\,\mathrm{d}u = -S(t)$,

$$m'(t) = \frac{-S(t)\cdot S(t) - \big(\int_t^\infty S(u)\,\mathrm{d}u\big)S'(t)}{S(t)^2}.$$

Since $S'(t) = -f(t) = -h(t)S(t)$,

$$m'(t) = \frac{-S(t)^2 + \big(\int_t^\infty S(u)\,\mathrm{d}u\big)h(t)S(t)}{S(t)^2} = -1 + h(t)\frac{\int_t^\infty S(u)\,\mathrm{d}u}{S(t)} = -1 + h(t)m(t).$$

■

**(2) Monotonicity [Supplement].** By (1), $m'(t) \leq 0 \iff h(t)m(t) \leq 1$ and $m'(t) \geq 0 \iff h(t)m(t) \geq 1$. Substituting $u \mapsto t + u$ and using $S(t+u)/S(t) = \exp\{-(H(t+u) - H(t))\}$,

$$m(t) = \int_0^\infty \frac{S(t+u)}{S(t)}\,\mathrm{d}u = \int_0^\infty \exp\Big(-\int_t^{t+u} h(s)\,\mathrm{d}s\Big)\,\mathrm{d}u.$$

- If $h$ is increasing, $h(s) \geq h(t)$ on $[t, t+u]$, so $\int_t^{t+u} h(s)\,\mathrm{d}s \geq u\,h(t)$ and

$$m(t) \leq \int_0^\infty e^{-uh(t)}\,\mathrm{d}u = \frac{1}{h(t)} \;\Rightarrow\; h(t)m(t) \leq 1 \;\Rightarrow\; m'(t) \leq 0.$$

- If $h$ is decreasing, $\int_t^{t+u} h(s)\,\mathrm{d}s \leq u\,h(t)$, so $m(t) \geq 1/h(t)$, $h(t)m(t) \geq 1$, and $m'(t) \geq 0$.

■

**Example 2.20: Application to the Weibull**

$h(t) = \lambda\alpha t^{\alpha-1}$.

- $\alpha < 1$: $h$ decreasing $\Rightarrow$ $m$ increasing.
- $\alpha = 1$: $h \equiv \lambda \Rightarrow m'(t) = \lambda m(t) - 1$; the bounded solution is $m \equiv 1/\lambda$ (so $m' = 0$).
- $\alpha > 1$: $h$ increasing $\Rightarrow$ $m$ decreasing.

# Lecture 3. Basic Parametric Distributions and Models

## Contents

## 1 Outline

- Basic parametric distributions for survival data: exponential, Weibull, extreme value, log-normal, log-logistic, gamma.
- Popular regression models for survival data: accelerated failure time (log-linear), multiplicative (proportional hazards), additive hazards.

## 2 Exponential Distribution

**Definition 3.1: Exponential($\lambda$)**

$f(x) = \lambda e^{-\lambda x}$, $S(x) = e^{-\lambda x}$, $h(x) = \lambda$ (constant), $H(x) = \lambda x$.

- The conditional probability of failure at any time $t$, given no event before $t$, does not depend on $t$ (memoryless).
- $H(x) = -\ln S(x) = \lambda x$ gives an **empirical check**: plot an estimate of $H(x)$ against $x$; a straight line through the origin supports the exponential model.
- Mathematically simple but too restrictive in both health and industrial applications.
- Special case of both the Weibull ($\alpha = 1$) and the gamma ($\beta = 1$).

## 3 Weibull Distribution

**Definition 3.2: Weibull($\lambda, \alpha$), $\lambda, \alpha > 0$**

$$h(x) = \lambda\alpha x^{\alpha-1}, \qquad H(x) = \lambda x^{\alpha}, \qquad S(x) = \exp(-\lambda x^{\alpha}).$$

- Mathematically simple, yet fairly flexible: increasing hazard ($\alpha > 1$), decreasing ($\alpha < 1$), constant ($\alpha = 1$).
- **Empirical check**: $\ln H(x) = \ln\lambda + \alpha \ln x$, so plot $\ln[\hat{H}(x)]$ against $\ln x$ and look for linearity (slope $\alpha$, intercept $\ln\lambda$).

## 4 Weibull and the Extreme Value Distribution

**Motivation.** The exponential model has constant hazard, which is rarely realistic. The Weibull distribution adds a shape parameter so the hazard can increase or decrease, and on the log scale it becomes a location–scale model, the basis of accelerated failure time regression.

**Definition 3.3: Standard extreme value distribution**

$$f_W(w) = \exp\{w - e^w\}, \qquad S_W(w) = \exp\{-e^w\}, \qquad -\infty < w < \infty.$$

It is a special case of the Gompertz family.

**Theorem 3.4: Log-linear representation of the Weibull**

If $Y \equiv \ln X = \mu + \sigma W$ with $W$ standard extreme value, then $X \sim \text{Weibull}(\lambda, \alpha)$ with

$$\alpha = \frac{1}{\sigma}, \qquad \lambda = e^{-\mu/\sigma} \quad \Big(\text{equivalently } \mu = \frac{-\ln \lambda}{\alpha},\ \sigma = \frac{1}{\alpha}\Big).$$

**Proof [Supplement].** First, $S_W(w) = \int_w^\infty e^{v-e^v}\,dv$; substituting $u = e^v$ gives $\int_{e^w}^\infty e^{-u}\,du = \exp(-e^w)$. Then

$$S_X(x) = P(\ln X > \ln x) = P\Big(W > \frac{\ln x - \mu}{\sigma}\Big) = \exp\Big\{-e^{(\ln x - \mu)/\sigma}\Big\} = \exp\Big\{-e^{-\mu/\sigma}x^{1/\sigma}\Big\},$$

which is $\exp(-\lambda x^\alpha)$ with $\alpha = 1/\sigma$, $\lambda = e^{-\mu/\sigma}$. ■

Hence, if the error in a log-linear model is standard extreme value, $X$ is Weibull.

## 5 Log-normal Distribution

**Definition 3.5**

If $Y = \ln X \sim N(\mu, \sigma^2)$ then $X \sim$ log-normal$(\mu, \sigma^2)$: $S(x) = 1 - \Phi\Big(\frac{\ln x - \mu}{\sigma}\Big)$.

- Close tie to the normal distribution.
- Can fit unimodal (hump-shaped) hazards: the hazard starts at 0, rises to a maximum and then decreases.
- Criticized as a lifetime distribution because the hazard *always decreases at the tail* (to 0).
- May still fit cases where large values of $x$ are not of interest.

## 6 Log-logistic Distribution

**Motivation.** Some hazards rise and then fall, for example after surgery. The log-logistic distribution allows this and has closed-form survival and hazard functions.

**Definition 3.6: Log-logistic**

If $Y = \ln X = \mu + \sigma W$ with $W$ standard logistic, LG$(0, 1)$:

$$f_W(w) = \frac{e^w}{(1+e^w)^2}, \qquad S_W(w) = \frac{1}{1+e^w},$$

then $Y \sim \text{LG}(\mu, \sigma^2)$ with $f_Y(y) = \dfrac{e^{(y-\mu)/\sigma}}{\sigma\big(1+e^{(y-\mu)/\sigma}\big)^2}$ and $X \sim$ log-logistic.

**Derivation: survival function of $X$**

$$S_X(x) = S_W\Big(\frac{\ln x - \mu}{\sigma}\Big) = \frac{1}{1+\exp\left(\frac{\ln x-\mu}{\sigma}\right)} = \frac{1}{1+e^{-\mu/\sigma}x^{1/\sigma}} = \frac{1}{1+\lambda x^{\alpha}},$$

with $\alpha = 1/\sigma$ and $\lambda = e^{-\mu/\sigma}$.

**Theorem 3.7: Log-logistic hazard**

$$h_X(x) = \frac{\lambda\alpha x^{\alpha-1}}{1+\lambda x^{\alpha}}.$$

It is monotone decreasing when $\alpha \le 1$. When $\alpha > 1$ it increases to a maximum at $x^* = \{(\alpha-1)/\lambda\}^{1/\alpha}$ and then decreases to 0 as $x \to \infty$.

**Proof [Supplement].** $H(x) = -\ln S(x) = \ln(1+\lambda x^{\alpha})$, so $h(x) = H'(x) = \lambda\alpha x^{\alpha-1}/(1+\lambda x^{\alpha})$. Differentiating,

$$h'(x) = \lambda\alpha\,\frac{(\alpha-1)x^{\alpha-2}(1+\lambda x^{\alpha}) - x^{\alpha-1}\lambda\alpha x^{\alpha-1}}{(1+\lambda x^{\alpha})^2} = \lambda\alpha\,\frac{x^{\alpha-2}\left[(\alpha-1)-\lambda x^{\alpha}\right]}{(1+\lambda x^{\alpha})^2}.$$

If $\alpha \le 1$ the bracket is negative for all $x > 0$. If $\alpha > 1$ the bracket is positive for $\lambda x^{\alpha} < \alpha - 1$ and negative afterwards, giving the maximum at $x^*$; and $h(x) \sim \alpha/x \to 0$. ■

The hazard resembles the log-normal one but has a closed form.

# 7 Gamma Distribution

**Definition 3.8: Gamma$(\lambda, \beta)$, $\lambda, \beta > 0$**

$$f(x) = \frac{\lambda^{\beta}x^{\beta-1}e^{-\lambda x}}{\Gamma(\beta)}, \qquad S(x) = \int_x^{\infty}\frac{\lambda^{\beta}s^{\beta-1}e^{-\lambda s}}{\Gamma(\beta)}\,\mathrm{d}s, \qquad h(x) = \frac{x^{\beta-1}e^{-\lambda x}}{\int_x^{\infty}s^{\beta-1}e^{-\lambda s}\,\mathrm{d}s}.$$

No closed form for $S$ or $h$ in general.

- $\beta = 1$: exponential$(\lambda)$. $\beta \to \infty$: approximately normal. $\lambda = 1/2$, integer $\beta$: $\chi^2_{2\beta}$.
- $\beta > 1$: $h$ is monotone increasing with $h(0) = 0$ and $h(x) \to \lambda$ as $x \to \infty$.
- $\beta < 1$: $h$ is monotone decreasing with $h(0) = \infty$ and $h(x) \to \lambda$ as $x \to \infty$.
- Not as mathematically tractable as the others; not widely used (the Weibull is more popular).

**Derivation: why the gamma hazard tends to $\lambda$ monotonically**

Substitute $s = x + u$ in the reciprocal hazard:

$$\frac{1}{h(x)} = \int_x^{\infty}\Big(\frac{s}{x}\Big)^{\beta-1}e^{-\lambda(s-x)}\,\mathrm{d}s = \int_0^{\infty}\Big(1+\frac{u}{x}\Big)^{\beta-1}e^{-\lambda u}\,\mathrm{d}u.$$

As $x \to \infty$, $(1+u/x)^{\beta-1} \to 1$, so $1/h(x) \to \int_0^{\infty}e^{-\lambda u}\,\mathrm{d}u = 1/\lambda$. For $\beta > 1$, $(1+u/x)^{\beta-1}$ decreases in $x$, so $1/h$ decreases and $h$ increases; for $\beta < 1$ the reverse holds.

# 8 Regression Models for Survival Data

To deal with heterogeneity in the population, the quantities summarizing survival must be adjusted for covariates (explanatory variables). For example, in non-randomized breast cancer studies the two groups may be imbalanced in age, tumour size, number of positive nodes, or hormonal status.

## 8.1 Approach 1: Log-linear / accelerated failure time (AFT) model

**Definition 3.9: Log-linear model**

$$Y \equiv \ln X = \mu + \boldsymbol{\gamma}'\mathbf{z} + \sigma W,$$

where $W \sim F$ (a known distribution) and $\mu, \sigma, \boldsymbol{\gamma}$ are unknown.

**Theorem 3.10: Log-linear ⇒ AFT**

Let $X_0 = \exp(\mu + \sigma W)$ with survival $S_0$. Then

$$S(x \mid \mathbf{z}) = S_0(x\, e^{-\boldsymbol{\gamma}'\mathbf{z}}), \qquad h(x \mid \mathbf{z}) = \eta\, h_0(\eta x), \quad \eta = e^{-\boldsymbol{\gamma}'\mathbf{z}}.$$

**Proof [Supplement].**

$$\begin{aligned} P(X > x \mid \mathbf{z}) &= P(Y > \ln x \mid \mathbf{z}) = P(\mu + \sigma W > \ln x - \boldsymbol{\gamma}'\mathbf{z}) \\ &= P\big(\exp(\mu + \sigma W) > x \exp(-\boldsymbol{\gamma}'\mathbf{z})\big) \\ &= S_0(xe^{-\boldsymbol{\gamma}'\mathbf{z}}). \end{aligned}$$

For the hazard, $S(x \mid \mathbf{z}) = S_0(\eta x)$ gives $f(x \mid \mathbf{z}) = \eta f_0(\eta x)$ (chain rule), so $h(x \mid \mathbf{z}) = \eta f_0(\eta x)/S_0(\eta x) = \eta h_0(\eta x)$. ■

- The covariate rescales time: time is **accelerated** if $\eta = e^{-\boldsymbol{\gamma}'\mathbf{z}} > 1$ and **decelerated** if $\eta < 1$.
- The log-linear model induces the AFT model but with **opposite signs** for the regression coefficients: a positive $\gamma$ lengthens survival ($\eta < 1$).

**Derivation: interpretation of $\gamma$**

For covariate values $\mathbf{z}_1, \mathbf{z}_2$, since $X = e^{\mu + \boldsymbol{\gamma}'\mathbf{z}} e^{\sigma W}$,

$$\frac{\mathrm{E}(X \mid \mathbf{z}_1)}{\mathrm{E}(X \mid \mathbf{z}_2)} = \frac{e^{\mu + \boldsymbol{\gamma}'\mathbf{z}_1}\mathrm{E}(e^{\sigma W})}{e^{\mu + \boldsymbol{\gamma}'\mathbf{z}_2}\mathrm{E}(e^{\sigma W})} = e^{\boldsymbol{\gamma}'(\mathbf{z}_1 - \mathbf{z}_2)} \implies \ln \frac{\mathrm{E}(X \mid \mathbf{z}_1)}{\mathrm{E}(X \mid \mathbf{z}_2)} = \boldsymbol{\gamma}'(\mathbf{z}_1 - \mathbf{z}_2).$$

So $\gamma$ is the log of the ratio of mean survival times for a unit increase in $z$. (The same holds for any quantile.)

**Definition 3.11: Three choices of $F$**

1. $W \sim N(0,1)$: $\ln X \sim N(\mu + \boldsymbol{\gamma}'\mathbf{z}, \sigma^2)$, i.e. $X$ is **log-normal**.
2. $W$ standard extreme value, $f_W(w) = \exp(w - e^w)$: $X \sim \text{Weibull}(\lambda, \alpha)$ with $\alpha = 1/\sigma$ and $\lambda = \exp\{-(\frac{\mu}{\sigma} + \frac{\boldsymbol{\gamma}'\mathbf{z}}{\sigma})\}$.
3. $W$ standard logistic: the **log-logistic** regression model.

## 8.2 Approach 2: Multiplicative (proportional hazards) model

**Definition 3.12: Multiplicative hazards model**

$$h(x \mid \mathbf{z}) = h_0(x)\, C(\boldsymbol{\beta}'\mathbf{z}),$$

where the baseline hazard $h_0(x) \geq 0$ may be parametric or left unspecified. With $C(\boldsymbol{\beta}'\mathbf{z}) = e^{\boldsymbol{\beta}'\mathbf{z}}$ this is the **Cox proportional hazards model**.

- $\beta$ is the **log hazard ratio**, constant over time: $h(x \mid \mathbf{z}_1)/h(x \mid \mathbf{z}_2) = e^{\boldsymbol{\beta}'(\mathbf{z}_1 - \mathbf{z}_2)}$.
- Integrating, $H(x \mid \mathbf{z}) = e^{\boldsymbol{\beta}'\mathbf{z}} H_0(x)$ and $S(x \mid \mathbf{z}) = \exp\{-e^{\boldsymbol{\beta}'\mathbf{z}} H_0(x)\} = \{S_0(x)\}^{e^{\boldsymbol{\beta}'\mathbf{z}}}$.
- **Graphical check** of PH: $\ln[-\ln S(x \mid \mathbf{z})] = \boldsymbol{\beta}'\mathbf{z} + \ln[-\ln S_0(x)]$, so log-cumulative-hazard curves for different $\mathbf{z}$ should be *parallel* (vertical shift $\boldsymbol{\beta}'\mathbf{z}$).

**Example 3.13: Weibull baseline under PH**

If $h_0(x) = \alpha\lambda x^{\alpha-1}$ then $h(x \mid \mathbf{z}) = \alpha\lambda x^{\alpha-1}\exp(\boldsymbol{\beta}'\mathbf{z})$ and

$$S(x \mid \mathbf{z}) = \exp\big[-\lambda \exp(\boldsymbol{\beta}'\mathbf{z}) x^{\alpha}\big], \quad \text{i.e.} \quad X \mid \mathbf{z} \sim \text{Weibull}(\lambda e^{\boldsymbol{\beta}'\mathbf{z}}, \alpha).$$

**Theorem 3.14: Weibull is closed under both AFT and PH**

The Weibull is the only continuous distribution that is closed under both the AFT and PH models.

**Derivation: matching the two parametrizations**

Under AFT with Weibull baseline, $h(x \mid \mathbf{z}) = \eta h_0(\eta x) = \eta\,\alpha\lambda(\eta x)^{\alpha-1} = \eta^{\alpha}\cdot\alpha\lambda x^{\alpha-1}$, a PH model with hazard ratio $\eta^{\alpha} = e^{-\alpha\boldsymbol{\gamma}'\mathbf{z}}$. Hence

$$\boldsymbol{\beta} = -\alpha\boldsymbol{\gamma} = -\boldsymbol{\gamma}/\sigma.$$

## 8.3 Approach 3: Additive hazards model

**Definition 3.15: Additive hazards model**

$$h(x \mid \mathbf{z}) = h_0(x) + \sum_{j=1}^{p} z_j(x)\beta_j(x).$$

- Allows covariate effects to vary over time.
- The regression functions may be positive or negative, so they must be constrained to keep $h(x \mid \mathbf{z}) \geq 0$.

# Lecture 4. Formal Introduction to Censoring and Truncation

## Contents

## 1 Right Censoring

**Definition 4.1: Right-censored data**

An observation is right censored if it is not known exactly, only that it is greater than some value. Let $X$ = time to the event of interest and $C$ = censoring time. We observe

$$(T, \delta), \qquad T = \min(X, C), \qquad \delta = I(X < C) \text{ (or } I(X \leq C)),$$

and the data are $n$ realizations $\{(T_i, \delta_i),\ i = 1, \ldots, n\}$.

There are many types of right censoring; some of them provide information on $C$ even when $X < C$.

### 1.1 Type I censoring

The event is observed only if it occurs before some *pre-specified* time.

- **Fixed Type I censoring**: $C_i = \tau$ for all $i$. Typical of controlled experiments, e.g. a mice study of time to death after a fixed period following intense carcinogen exposure: every animal still alive at the end of the study is censored at the same time.
- **Progressive Type I censoring**: $C$ is fixed in advance but may take different values for each $i$, with only a finite number of distinct values. *Example*: $X$ = time to a certain % weight loss, weights measured every 12 hours; mice in cages 1–2 are sacrificed at the end of week 1 and those in cages 3–4 by the end of week 2. (A second example uses two sacrifice times, 42 and 104 weeks.)
- **Generalized Type I censoring**: $C$ may differ for every individual but is known in advance. Common in human studies with *staggered entry* and complete follow-up to the end of the study, e.g. a Phase I chemotherapy study with $X$ = time to toxicity: $C_i$ = (end of study) − (entry time of $i$) is computable at enrolment, assuming a fixed study length and no dropout.

### 1.2 Type II censoring

- **Simple Type II**: sample size $n$; the study continues until the first $r$ failures. The number of failures $r$ and censored observations $n - r$ are *fixed*; the censoring time $C = X_{(r)}$ is *random*.
- **Progressive Type II**: after the first $r_1$ failures, $n_1 - r_1$ of the remaining $n - r_1$ unfailed items are removed (sacrificed); after the next $r_2$ failures, $n_2 - r_2$ of the unfailed items are removed; and so on until a pre-decided number of repetitions is completed.

**Remark 4.2: Type I vs. Type II**

Type I: $C$ is fixed (predetermined before the study starts). Type II: $C$ is random, namely the time at which a predetermined number of events has occurred. This distinction matters when constructing the likelihood.

### 1.3 Random censoring

Typical of human studies: $C \sim G(t)$, not fixed in advance and possibly unknown.

- Scenario 1: censoring at the end of study ("administrative loss to follow-up"), fixed in advance, similar to Type I.
- Scenario 2: censoring before the end of study ("drop-out"), random, due to other factors.
- In practice censoring is a mixture of the two.

## 2 Left, Double and Interval Censoring

**Definition 4.3: Left censoring**

An observation is left censored if it is not known exactly, only that it is less than some value $C_l$. The exact lifetime is known iff $X \geq C_l$. Data: $(T, \delta)$ with $T = \max(X, C_l)$, $\delta = I(X \geq C_l)$.

**Definition 4.4: Double censoring**

With left censoring time $C_l$ and right censoring time $C_r$ $(C_l < C_r)$:

$$T = \max\{\min(X, C_r),\, C_l\}, \qquad \delta = \begin{cases} 1, & C_l \leq X \leq C_r \text{ (exact)} \\ -1, & X < C_l \text{ (left censored)} \\ 0, & X > C_r \text{ (right censored)} \end{cases}$$

**Remark 4.5: Double vs. interval censoring**

**Interval censoring**: each observation is only known to lie in an interval, $L < X \leq R$; no exact time is observed. **Double censoring**: each observation is one of three possibilities — fully observed, right censored, or left censored.

## 3 Truncation

**Definition 4.6: Truncation**

Only individuals whose event time lies within an observational window $(Y_L, Y_R)$ are observed.

- $Y_R = \infty$: **left truncation**; we observe only those with $Y_L < X$.
- $Y_L = 0$: **right truncation**; we observe only those with $X \leq Y_R$.

Inference for truncated data is restricted to *conditional* estimation.

**Example 4.7: Breast cancer onset (classification exercise)**

To estimate the distribution of age at breast cancer onset in post-menopausal women, 8 healthy 50-year-old women had yearly mammograms for 10 years; no tumours were detected between scheduled exams. Onset ages:

$$(55, 56],\ (58, 59],\ (52, 53],\ (59, 60],\ > 60,\ > 60,\ > 60,\ > 60.$$

**Answer.** The four detected cases are **interval censored** (onset between two yearly exams). The four undetected cases are **right censored** at 60. Moreover, only women who were cancer-free at age 50 were enrolled, so the ages are **left truncated** at 50 (inference is conditional on $X > 50$).

# 4 Likelihood Construction

## 4.1 Contributions of each observation type

Let $X$ be continuous with pdf $f$, cdf $F$ and $S = 1 - F$; $C_r$ = right censoring point, $C_l$ = left censoring point, $(L, R]$ = censoring interval, $Y$ = truncation point.

| Observation | Likelihood contribution |
|---|---|
| Exact lifetime | $f(x)$ |
| Right censored | $S(C_r) = P(X > C_r)$ |
| Left censored | $1 - S(C_l) = P(X < C_l)$ |
| Interval censored | $S(L) - S(R) = P(L < X \leq R)$ |
| Left truncated | $f(x)/S(Y_L)$ |
| Right truncated | $f(x)/[1 - S(Y_R)]$ |
| Interval truncated | $f(x)/[S(Y_L) - S(Y_R)]$ |

The truncated contributions are conditional densities: e.g. for left truncation, $f(x \mid X > Y_L) = f(x)/P(X > Y_L) = f(x)/S(Y_L)$.

**Theorem 4.8: General censored-data likelihood**

$$L \propto \prod_{i \in D} f(x_i) \prod_{i \in RC} S(C_{r,i}) \prod_{i \in LC} F(C_{l,i}) \prod_{i \in IC} \{S(L_i) - S(R_i)\},$$

where $D$ = exact (death) times, $RC/LC/IC$ = right-/left-/interval-censored observations.

**Truncation.** For left-truncated data (possibly also right censored), replace $f(x_i)$ by $f(x_i)/S(Y_i)$ and $S(C_{r,i})$ by $S(C_{r,i})/S(Y_i)$. For right-truncated data (only events observed), replace $f(x_i)$ by $f(x_i)/[1 - S(Y_i)]$.

## 4.2 Type I right censoring

**Theorem 4.9**

Observing $(T_i, \delta_i)$ with $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and fixed $C_i$,

$$L = \prod_{i=1}^{n} \Pr(t_i, \delta_i) = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i} = \prod_{i=1}^{n} h(t_i)^{\delta_i} e^{-H(t_i)}.$$

**Proof [Supplement].** Since $C_i$ is fixed, $\delta_i = 0$ means $T_i = C_i$ and $X_i > C_i$, with probability $S(C_i) = S(t_i)$; $\delta_i = 1$ means $T_i = X_i \leq C_i$ with density $f(t_i)$. For the last form, $f = hS$ and $S = e^{-H}$: $f^{\delta} S^{1-\delta} = (hS)^{\delta} S^{1-\delta} = h^{\delta} S = h^{\delta} e^{-H}$. ■

**Example 4.10: Exponential with right censoring**

$X \sim \text{Exp}(\lambda)$, $T = \min(X, C)$, $\delta = I(X < C)$:

| $T_i$ | 0.5 | 1 | 0.75 | 0.25 | 0.75 |
|---|---|---|---|---|---|
| $\delta_i$ | 1 | 1 | 1 | 0 | 0 |

With $h(t) = \lambda$ and $H(t) = \lambda t$,

$$L(\lambda) = \prod_{i=1}^{5} \lambda^{\delta_i} e^{-\lambda t_i} = \lambda^{\sum \delta_i} e^{-\lambda \sum t_i} = \lambda^3 e^{-3.25\lambda}.$$

Maximizing $\ell(\lambda) = 3\ln\lambda - 3.25\lambda$: $\ell'(\lambda) = 3/\lambda - 3.25 = 0 \Rightarrow \hat{\lambda} = 3/3.25 \approx 0.923$. In general $\hat{\lambda} = \sum \delta_i / \sum t_i$ = (number of events)/(total time at risk).

## 4.3 Type II censoring

**Theorem 4.11**

With $n$ i.i.d. $X_i \sim f$, observing the $r$ smallest order statistics $X_{(1)} < \cdots < X_{(r)}$ and $n - r$ values censored at $X_{(r)}$,

$$L = \frac{n!}{(n-r)!} \prod_{i=1}^{r} f(x_{(i)}) \left\{S(x_{(r)})\right\}^{n-r}.$$

**Derivation: where the constant comes from**

The joint density of the first $r$ order statistics: choose which $r$ of the $n$ units fail (and in which order) — $n!/(n-r)!$ ordered choices — each failed unit contributes $f(x_{(i)})$, and the remaining $n - r$ units must all survive beyond $x_{(r)}$, contributing $S(x_{(r)})^{n-r}$. The constant does not involve the parameters, so the kernel is the same as in Theorem 4.8.

**Progressive Type II** (two stages): first $r_1$ ordered failures $X_{(1)}, \ldots, X_{(r_1)}$, then $n_1 - r_1$ items removed; the next $r_2$ failures $X^*_{(1)}, \ldots, X^*_{(r_2)}$, then the remaining $n - n_1 - r_2$ removed:

$$L \propto \prod_{i=1}^{r_1} f(x_{(i)}) \left\{S(x_{(r_1)})\right\}^{n_1 - r_1} \prod_{i=1}^{r_2} f(x^*_{(i)}) \left\{S(x^*_{(r_2)})\right\}^{n - n_1 - r_2} \propto \prod_{i \in D} f(x_i) \prod_{i \in RC} S(C_i).$$

## 4.4 Random censoring

$X_i \sim f, S$ and $C_i \sim g, G$ are both random but **independent**; we observe $T_i = \min(X_i, C_i)$ and $\delta_i = I(X_i \le C_i)$.

**Theorem 4.12**

$$L = \prod_{i=1}^{n} \left\{f(t_i)G(t_i)\right\}^{\delta_i} \left\{g(t_i)S(t_i)\right\}^{1-\delta_i} = \underbrace{\prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i}}_{L_X} \underbrace{\prod_{i=1}^{n} g(t_i)^{1-\delta_i} G(t_i)^{\delta_i}}_{L_C}.$$

**Proof [Supplement].** For $\delta = 1$: $P(T \in \mathrm{d}t, \delta = 1) = P(X \in \mathrm{d}t,\ C \ge t) = f(t)G(t)\,\mathrm{d}t$ by independence. For $\delta = 0$: $P(T \in \mathrm{d}t, \delta = 0) = P(C \in \mathrm{d}t,\ X > t) = g(t)S(t)\,\mathrm{d}t$. Multiply over $i$ and regroup. ■

If $f = f(\cdot;\theta)$ and the distribution of $C$ does not depend on $\theta$ (**non-informative censoring**), $L_C$ is constant in $\theta$ and

$$L \propto L_X = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i},$$

the same form as under Type I censoring.

## 5 Remarks

- These likelihoods are used primarily for fitting parametric models (K&M Ch. 12).
- They also underlie the *partial likelihood* of the semiparametric Cox model (K&M Ch. 8–9).
- In most applications they are not used explicitly, but the rationale — how each data component contributes, as in Theorem 4.8 — is essential.

# Lecture 5. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 1)

## Contents

## 1 Setting

$X$ is a continuous failure time with survival function $S(t)$ and cumulative hazard $H(t)$. $X$ is subject to **non-informative right censoring**, so we observe

$$\{(T_i, \delta_i),\ i = 1, \ldots, n\}, \qquad T_i = \min(X_i, C_i), \quad \delta_i = I(X_i \le C_i).$$

Goal: estimate $S(t)$ and $H(t)$ without assuming a parametric family. We build up the estimator in four steps: discrete/continuous × without/with censoring.

## 2 Building Up the Estimator

### 2.1 Discrete $X$ without censoring

Let $X$ have $M$ support points $0 < \tilde{x}_1 < \cdots < \tilde{x}_M < \infty$. With $\tilde{x}_0 = 0$, $S(\tilde{x}_0) = 1$, the product-limit representation (Lecture 2) is

$$\begin{aligned} S(x) = P(X > x) &= \prod_{\tilde{x}_j \le x} \frac{S(\tilde{x}_j)}{S(\tilde{x}_{j-1})} = \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X > \tilde{x}_{j-1}) \\ &= \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X \ge \tilde{x}_j) = \prod_{\tilde{x}_j \le x} [1 - h(\tilde{x}_j)], \qquad h(\tilde{x}_j) = P(X = \tilde{x}_j \mid X \ge \tilde{x}_j). \end{aligned}$$

(The third equality holds because there is no mass strictly between $\tilde{x}_{j-1}$ and $\tilde{x}_j$.) With i.i.d. $X_1, \ldots, X_n$, estimate

$$\hat{h}(\tilde{x}_j) = \frac{\#\{X_i = \tilde{x}_j\}}{\#\{X_i \ge \tilde{x}_j\}} = \frac{\#\text{ deaths at } \tilde{x}_j}{\#\text{ deaths at } \tilde{x}_j, \tilde{x}_{j+1}, \ldots, \tilde{x}_M}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \le x} [1 - \hat{h}(\tilde{x}_j)].$$

$\hat{h}$ and $\hat{S}$ are the MLEs of $h$ and $S$.

### 2.2 Discrete $X$ with right censoring

With $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and $X_i \perp\!\!\!\perp C_i$:

$$\begin{aligned} P(T_i = \tilde{x}_j, \delta_i = 1 \mid T_i \ge \tilde{x}_j) &= \frac{P(T_i = \tilde{x}_j, \delta_i = 1)}{P(T_i \ge \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j,\ C_i \ge \tilde{x}_j)}{P(X_i \ge \tilde{x}_j,\ C_i \ge \tilde{x}_j)} \\ &= \frac{P(X_i = \tilde{x}_j)P(C_i \ge \tilde{x}_j)}{P(X_i \ge \tilde{x}_j)P(C_i \ge \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j)}{P(X_i \ge \tilde{x}_j)} = h(\tilde{x}_j). \end{aligned}$$

So the hazard is identifiable from the *observed* data, which suggests

$$\hat{h}(\tilde{x}_j) = \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \ge \tilde{x}_j\}}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \le x} \Big[1 - \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \ge \tilde{x}_j\}}\Big].$$

**Derivation: $\hat{h}$ is the MLE under censoring**

Under independent censoring, the likelihood in the hazard parameters factorizes over support points. With $d_j$ deaths among $Y_j$ at risk at $\tilde{x}_j$, each subject at risk either dies at $\tilde{x}_j$ (prob. $h_j$) or survives past it (prob. $1-h_j$); censored subjects contribute survival up to their censoring time. Thus

$$L(h_1,\dots,h_M) \propto \prod_j h_j^{d_j}(1-h_j)^{Y_j-d_j}, \qquad \frac{\partial \log L}{\partial h_j} = \frac{d_j}{h_j} - \frac{Y_j-d_j}{1-h_j} = 0 \;\Rightarrow\; \hat{h}_j = \frac{d_j}{Y_j}.$$

By invariance, $\hat{S}(x) = \prod_{\tilde{x}_j \le x}(1-\hat{h}_j)$ is the MLE of $S$.

### 2.3 Continuous $X$

Without censoring, the survival function is smooth and can be estimated by the complement of the empirical distribution function:

$$\hat{S}(t) = 1 - \mathbb{F}_n(t) = 1 - \frac{1}{n}\sum_{i=1}^{n} I(X_i \le t).$$

With right censoring we must modify this. Two questions guide the construction: *Should the estimator jump at censoring times?* (No — no event is observed there.) *What is the contribution of a censored observation?* (It counts in the risk set $Y_j$ for every death time up to its censoring time, then leaves.)

## 3 The Kaplan–Meier (Product-Limit) Estimator

**Motivation.** With censored data the empirical survival function cannot be computed, since we do not know when censored subjects fail. Kaplan–Meier writes survival as a product of conditional survival probabilities over the observed failure times, each estimated from those still at risk.

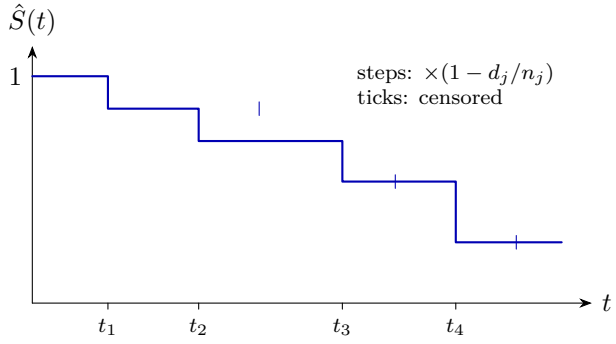



**Figure.** *The Kaplan–Meier estimate is a step function that drops only at observed failure times $t_j$, by the factor $1-d_j/n_j$. Censored observations (ticks) cause no drop but leave the risk set, which makes later drops larger.*

**Definition 5.1: Notation**

- $t_1 < t_2 < \cdots < t_D$: the $D$ distinct death times;
- $d_j = \sum_{i=1}^n I(T_i = t_j, \delta_i = 1)$: number of deaths at $t_j$;
- $Y_j = \sum_{i=1}^n I(T_i \ge t_j)$: number at risk (alive and under observation) just before $t_j$.

**Theorem 5.2: Kaplan–Meier estimator**

$$\hat{S}(t) = \begin{cases} 1, & t < t_1, \\ \prod_{t_j \le t} \left(1 - \frac{d_j}{Y_j}\right), & t_1 \le t. \end{cases}$$

**Remark 5.3: Ties between deaths and censorings**

If a death and a censoring occur at the same recorded time, the convention is that the censoring happens just *after* the death, so the censored subject is still counted in $Y_j$ ($Y_j$ uses $T_i \ge t_j$).

**Theorem 5.4: Greenwood's variance estimator**

$$\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)}.$$

**Derivation: delta-method derivation (Cox & Oakes, 1984)**

Let $\hat{h}_j = d_j/Y_j$. Treating $d_j \mid Y_j \sim \mathrm{Bin}(Y_j, h_j)$ and the $\hat{h}_j$ as approximately uncorrelated across $j$,

$$\mathrm{Var}(\hat{h}_j) \approx \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j}.$$

Since $\ln \hat{S}(t) = \sum_{t_j \le t} \ln(1 - \hat{h}_j)$ and, by the delta method with $g(u) = \ln(1-u)$, $g'(u) = -1/(1-u)$:

$$\begin{aligned} \mathrm{Var}\{\ln \hat{S}(t)\} &\approx \sum_{t_j \le t} \mathrm{Var}\{\ln(1-\hat{h}_j)\} \approx \sum_{t_j \le t} \left(\frac{1}{1-\hat{h}_j}\right)^2 \mathrm{Var}(\hat{h}_j) \\ &\approx \sum_{t_j \le t} \left(\frac{1}{1-\hat{h}_j}\right)^2 \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j} = \sum_{t_j \le t} \frac{\hat{h}_j}{Y_j(1-\hat{h}_j)} = \sum_{t_j \le t} \frac{d_j/Y_j}{Y_j(Y_j-d_j)/Y_j} \\ &= \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j-d_j)}. \end{aligned}$$

A second delta-method step with $\hat{S} = \exp(\ln \hat{S})$ gives $\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \,\mathrm{Var}\{\ln \hat{S}(t)\}$.

## 4 The Nelson–Aalen Estimator

**Motivation.** Estimating the cumulative hazard by summing the observed failure rates $d_i/n_i$ gives the Nelson–Aalen estimator, which has better small-sample properties and is the natural input for graphical model checks.

**Theorem 5.5: Nelson–Aalen estimator and Aalen's variance**

$$\tilde{H}(t) = \begin{cases} 0, & t < t_1, \\ \sum_{t_j \le t} \frac{d_j}{Y_j}, & t_1 \le t, \end{cases} \qquad \sigma_H^2(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

**Remark 5.6: Intuition**

$H(t) = \int_0^t h(s)\,ds$ accumulates the hazard; at each death time the hazard increment is estimated by $\hat{h}_j = d_j/Y_j$. Aalen's variance treats $d_j$ as approximately Poisson with mean $Y_j h_j$, so $\mathrm{Var}(d_j/Y_j) \approx Y_j h_j / Y_j^2 \approx d_j/Y_j^2$.

### 4.1 Alternative estimators

- An alternative estimator of $H$: $\hat{H}(t) = -\log \hat{S}(t)$.
- An alternative estimator of $S$: $\tilde{S}(t) = \exp\{-\tilde{H}(t)\}$ (the Breslow / Fleming–Harrington estimator).
- Since $1 - x \le e^{-x}$, each KM factor $1 - d_j/Y_j \le e^{-d_j/Y_j}$, so $\hat{S}(t) \le \tilde{S}(t)$ and equivalently $\tilde{H}(t) \le \hat{H}(t)$.
- $\hat{S}(t)$ is called the "product-limit" estimator: as $n \uparrow \infty$, $D \uparrow \infty$ and the number of terms in the product $\to \infty$ for continuous $X$. For finite $n$, $D < \infty$; in the limit the product runs over the whole support of $X$ (a *product integral*).

## 5 Worked Example (Small Sample)

**Example 5.7:** $n = 5$

Continuous $X$ subject to right censoring, $T = \min(X, C)$, $\delta = I(X \le C)$:

| $T_i$ | 0.5 | 1 | 0.75 | 0.25 | 0.75 |
|---|---|---|---|---|---|
| $\delta_i$ | 1 | 1 | 1 | 0 | 0 |

Ordered: $0.25^+$, 0.5, 0.75, $0.75^+$, 1. At $t = 0.75$ the censored subject is still at risk (tie convention).

| $t_j$ | $d_j$ | $Y_j$ | $1 - d_j/Y_j$ | $\hat{S}(t_j)$ | $d_j/Y_j$ | $\tilde{H}(t_j)$ | $\tilde{S}(t_j) = e^{-\tilde{H}}$ |
|---|---|---|---|---|---|---|---|
| 0.5 | 1 | 4 | 3/4 | 0.750 | 0.250 | 0.250 | 0.779 |
| 0.75 | 1 | 3 | 2/3 | 0.500 | 0.333 | 0.583 | 0.558 |
| 1 | 1 | 1 | 0 | 0.000 | 1.000 | 1.583 | 0.205 |

Greenwood: $\hat{V}[\hat{S}(0.5)] = 0.75^2 \cdot \frac{1}{4\cdot 3} = 0.0469$, $\hat{V}[\hat{S}(0.75)] = 0.5^2(\frac{1}{12} + \frac{1}{3\cdot 2}) = 0.0625$; at $t = 1$, $Y_j - d_j = 0$ and Greenwood is undefined (the KM drops to 0).

## 6 Remarks

- $\hat{S}$, $\tilde{H}$, $\hat{H}$ are right-continuous step functions even though $S$ and $H$ may be continuous.
- $\hat{S}$ and $\tilde{S}$ are consistent for $S(t)$ for all $t \le \tau$ such that $P(T \ge \tau) > 0$, and are asymptotically equivalent.
- $\hat{S}$ and $\tilde{H}$ are only defined for $t \le \max(T_i) = t_{\max}$. If the largest observation is censored, $\hat{S}$ does not reach 0. Some extensions for $t > t_{\max}$:

– Efron (1967): $\hat{S}(t) = 0$;
– Gill (1980): $\hat{S}(t) = \hat{S}(t_{\max})$;
– Brown, Hollander & Kowar (1974): $\hat{S}(t) = \exp\{t \log[\hat{S}(t_{\max})]/t_{\max}\}$ (an exponential tail).

# 7 The 6-MP Leukaemia Trial

**Data.** 42 children with acute leukaemia in complete or partial remission were matched in pairs by remission status and randomized within the pair to 6-mercaptopurine (6-MP) or placebo maintenance therapy, then followed until relapse or end of study. Times to relapse (weeks) for the 21 **6-MP** patients ($^+$ = censored):

$$6,\ 6,\ 6,\ 6^+,\ 7,\ 9^+,\ 10,\ 10^+,\ 11^+,\ 13,\ 16,\ 17^+,\ 19^+,\ 20^+,\ 22,\ 23,\ 25^+,\ 32^+,\ 32^+,\ 34^+,\ 35^+.$$

Placebo times (all observed): $1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17, 22, 23$.

**Example 5.8: Kaplan–Meier estimate for the 6-MP group**

| $t_j$ | $d_j$ | $Y_j$ | $\hat{S}(t) = \prod(1 - d_j/Y_j)$ | $\sum \frac{d_j}{Y_j(Y_j - d_j)}$ | $\hat{S}(t)^2 \sum \cdots$ |
|---|---|---|---|---|---|
| 6 | 3 | 21 | $[1 - \frac{3}{21}] = 0.857$ | $\frac{3}{21\times 18} = 0.0079$ | 0.0058 |
| 7 | 1 | 17 | $0.857(1 - \frac{1}{17}) = 0.807$ | $0.0079 + \frac{1}{17\times 16} = 0.0116$ | 0.0076 |
| 10 | 1 | 15 | $0.807(1 - \frac{1}{15}) = 0.753$ | $0.0116 + \frac{1}{15\times 14} = 0.0164$ | 0.0093 |
| 13 | 1 | 12 | $0.753(1 - \frac{1}{12}) = 0.690$ | $0.0164 + \frac{1}{12\times 11} = 0.0240$ | 0.0114 |
| 16 | 1 | 11 | $0.690(1 - \frac{1}{11}) = 0.628$ | $0.0240 + \frac{1}{11\times 10} = 0.0330$ | 0.0130 |
| 22 | 1 | 7 | $0.628(1 - \frac{1}{7}) = 0.538$ | $0.0330 + \frac{1}{7\times 6} = 0.0569$ | 0.0164 |
| 23 | 1 | 6 | $0.538(1 - \frac{1}{6}) = 0.448$ | $0.0569 + \frac{1}{6\times 5} = 0.0902$ | 0.0181 |

Note how censorings reduce $Y_j$ without producing a jump: e.g. between $t = 7$ ($Y = 17$) and $t = 10$ ($Y = 15$) the subject censored at $9^+$ leaves the risk set.

**Example 5.9: Nelson–Aalen estimate for the 6-MP group**

| $t$ | $\tilde{H}(t) = \sum d_j/Y_j$ | $\sigma_H^2 = \sum d_j/Y_j^2$ | SE |
|---|---|---|---|
| $0 \le t < 6$ | 0 | 0 | 0 |
| $6 \le t < 7$ | $3/21 = 0.1428$ | $3/21^2 = 0.0068$ | 0.0825 |
| $7 \le t < 10$ | $0.1428 + 1/17 = 0.2017$ | $0.0068 + 1/17^2 = 0.0103$ | 0.1015 |
| $10 \le t < 13$ | $0.2017 + 1/15 = 0.2683$ | $0.0103 + 1/15^2 = 0.0147$ | 0.1212 |
| $13 \le t < 16$ | $0.2683 + 1/12 = 0.3517$ | $0.0147 + 1/12^2 = 0.0217$ | 0.1473 |
| $16 \le t < 22$ | $0.3517 + 1/11 = 0.4426$ | $0.0217 + 1/11^2 = 0.0299$ | 0.1729 |
| $22 \le t < 23$ | $0.4426 + 1/7 = 0.5854$ | $0.0299 + 1/7^2 = 0.0503$ | 0.2243 |
| $23 \le t < 35$ | $0.5854 + 1/6 = 0.7521$ | $0.0503 + 1/6^2 = 0.0781$ | 0.2795 |

**Comparison.** Plotting $\tilde{S} = e^{-\tilde{H}}$ against $\hat{S}$ (K&M Fig. 4.1A), the Nelson–Aalen-based curve lies slightly *above* the KM curve at every step, and correspondingly $\tilde{H} = -\log \hat{S}$ lies slightly above $\tilde{H}$; the gap widens in the tail where $Y_j$ is small (e.g. at $t = 23$: $\hat{S} = 0.448$ vs. $\tilde{S} = e^{-0.752} = 0.471$).

# Lecture 6. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 2)

## Contents

## 1 Recap

$$\hat{S}(t) = \prod_{t_j \le t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)}; \qquad \tilde{H}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j}, \quad \sigma_H^2(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

Throughout, define

$$\sigma_S^2(t) = \frac{\hat{V}[\hat{S}(t)]}{\hat{S}^2(t)} = \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j - d_j)},$$

so that $\hat{S}(t)\sigma_S(t)$ is the Greenwood standard error.

## 2 Pointwise Confidence Intervals for $S(t)$

**Motivation.** An estimate of $S(t)$ needs an interval. Greenwood's variance gives one directly, but the interval may leave $[0, 1]$; transformations such as log and log-log fix this.

### 2.1 Linear interval

**Definition 6.1: Linear CI**

$$\Big[\hat{S}(t) - Z_{1-\alpha/2}\hat{S}(t)\sigma_S(t), \quad \hat{S}(t) + Z_{1-\alpha/2}\hat{S}(t)\sigma_S(t)\Big].$$

It is symmetric about $\hat{S}(t)$ and can fall outside $[0, 1]$.

### 2.2 Transformed intervals: the general recipe

**Derivation: delta method for a transformation $g$**

Let $g$ be known, monotone and differentiable. By the delta method, $\mathrm{SE}\{g(\hat{S})\} \approx |g'(\hat{S})| \cdot \mathrm{SE}(\hat{S}) = |g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t)$. A $100(1-\alpha)\%$ CI for $g(S(t))$ is

$$g(\hat{S}(t)) \pm Z_{1-\alpha/2}\,|g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t),$$

and re-transforming gives a CI for $S(t)$:

$$g^{-1}\Big\{g(\hat{S}(t)) \pm Z_{1-\alpha/2}\,|g'(\hat{S}(t))|\,\hat{S}(t)\sigma_S(t)\Big\}.$$

**Why transform?** Convergence to normality may be faster on the transformed scale, and the re-transformed CI respects $0 \le S(t) \le 1$.

### 2.3 Log-log transformation

**Theorem 6.2: Log-log CI**

$$\Big(\hat{S}(t)^{1/\theta},\ \hat{S}(t)^{\theta}\Big), \qquad \theta = \exp\Big\{\frac{Z_{1-\alpha/2}\,\sigma_S(t)}{\log \hat{S}(t)}\Big\}.$$

**Derivation [Supplement].** K&M use $\log H(t) = \log\{-\log S(t)\}$, i.e. $g(u) = \log\{-\log u\}$ with $g'(u) = \dfrac{1}{u \log u}$ and $g^{-1}(v) = \exp\{-\exp(v)\}$. Then

$$|g'(\hat{S})|\hat{S}\sigma_S = \frac{\sigma_S}{|\log \hat{S}|},$$

so the CI for $g(S)$ is $\log\{-\log \hat{S}\} \pm a$ with $a = Z_{1-\alpha/2}\sigma_S/|\log \hat{S}|$. Re-transforming,

$$g^{-1}\big(\log\{-\log \hat{S}\} \pm a\big) = \exp\{\log \hat{S} \cdot e^{\pm a}\} = \hat{S}^{e^{\pm a}}.$$

Since $\log \hat{S} < 0$, $\theta = e^{-a} < 1$; the endpoints are $\hat{S}^{1/\theta}$ (lower, as $1/\theta > 1$) and $\hat{S}^{\theta}$ (upper). ■

### 2.4 Arcsine-square-root transformation

**Theorem 6.3: Arcsine-square-root CI**

$$\text{lower} = \sin^2\Big\{\max\Big[0,\ \arcsin\big(\hat{S}(t)^{1/2}\big) - 0.5Z_{1-\alpha/2}\sigma_S(t)\Big(\frac{\hat{S}(t)}{1-\hat{S}(t)}\Big)^{1/2}\Big]\Big\},$$

$$\text{upper} = \sin^2\Big\{\min\Big[\frac{\pi}{2},\ \arcsin\big(\hat{S}(t)^{1/2}\big) + 0.5Z_{1-\alpha/2}\sigma_S(t)\Big(\frac{\hat{S}(t)}{1-\hat{S}(t)}\Big)^{1/2}\Big]\Big\}.$$

**Derivation [Supplement].** $g(u) = \arcsin(\sqrt{u})$ has $g'(u) = \dfrac{1}{\sqrt{1-u}} \cdot \dfrac{1}{2\sqrt{u}}$, so $|g'(\hat{S})|\hat{S}\sigma_S = \dfrac{\hat{S}\sigma_S}{2\sqrt{\hat{S}(1-\hat{S})}} = 0.5\,\sigma_S\Big(\dfrac{\hat{S}}{1-\hat{S}}\Big)^{1/2}$. The max / min keep the angle in $[0, \pi/2]$, and $g^{-1}(v) = \sin^2 v$. ■

### 2.5 Log transformation

**Definition 6.4: Log CI (default of R `survfit`)**

$g(S) = \log S$, $g'(u) = 1/u$, so $|g'(\hat{S})|\hat{S}\sigma_S = \sigma_S$:

$$\min\Big(\exp\big\{\log \hat{S}(t) \pm Z_{1-\alpha/2}\sigma_S(t)\big\},\ 1\Big).$$

Less recommended than log-log and arcsine-square-root.

**Remark 6.5: Comments**

- Log-log and arcsine-square-root CIs have better coverage than the linear CI.
- For very large samples the three methods are equivalent.

- The transformed CIs are *not* symmetric about the point estimate; they are appropriate in small samples where the point estimator is biased and its distribution skewed.

**Example 6.6: Toy example, $n = 7$ [Supplement]**

| $T_i$ | 1.5 | 2.5 | 1.4 | 6.2 | 2.8 | 5.3 | 4.5 |
|---|---|---|---|---|---|---|---|
| $\delta_i$ | 1 | 0 | 0 | 1 | 1 | 1 | 0 |

Ordered: $1.4^+$, 1.5, $2.5^+$, 2.8, $4.5^+$, 5.3, 6.2.

| $t_j$ | $d_j$ | $Y_j$ | $\hat{S}$ | $\sigma_S$ | Linear | Log-log | Arcsine |
|---|---|---|---|---|---|---|---|
| 1.5 | 1 | 6 | 0.833 | 0.183 | (0.535, 1.000*) | (0.273, 0.975) | (0.465, 1.000) |
| 2.8 | 1 | 4 | 0.625 | 0.342 | (0.207, 1.000*) | (0.142, 0.893) | (0.213, 0.949) |
| 5.3 | 1 | 2 | 0.313 | 0.785 | (0.000*, 0.794) | (0.013, 0.734) | (0.006, 0.804) |
| 6.2 | 1 | 1 | 0 | — | — | — | — |

*The raw linear limits (1.131, 1.043, −0.169) fall outside $[0, 1]$; transformed intervals stay inside automatically. Computation at $t = 2.8$: $\sigma_S^2 = \frac{1}{6\cdot5} + \frac{1}{4\cdot3} = 0.1167$, $\theta = \exp\{1.96 \times 0.3416/\ln 0.625\} = 0.241$, so the log-log CI is $(0.625^{1/0.241}, 0.625^{0.241}) = (0.142, 0.893)$.

# 3 Pointwise Confidence Intervals for $H(t)$

Based on the Nelson–Aalen $\tilde{H}(t)$ and $\sigma_H(t)$:

- **Linear**: $\tilde{H}(t) \pm Z_{1-\alpha/2}\sigma_H(t)$.
- **Log-transformed** (on $H$): $[\tilde{H}(t)/\phi,\ \phi\tilde{H}(t)]$, $\phi = \exp\left[\frac{Z_{1-\alpha/2}\sigma_H(t)}{\tilde{H}(t)}\right]$ (since $\mathrm{SE}(\log\tilde{H}) \approx \sigma_H/\tilde{H}$).
- **Arcsine-square-root** (on $S = e^{-H}$):

$$\Big[-2\ln\Big\{\sin\Big(\min\Big[\frac{\pi}{2},\ \arcsin\left(e^{-\tilde{H}(t)/2}\right) + 0.5Z_{1-\alpha/2}\sigma_H(t)\{e^{\tilde{H}(t)}-1\}^{-1/2}\Big]\Big)\Big\},$$
$$-2\ln\Big\{\sin\Big(\max\Big[0,\ \arcsin\left(e^{-\tilde{H}(t)/2}\right) - 0.5Z_{1-\alpha/2}\sigma_H(t)\{e^{\tilde{H}(t)}-1\}^{-1/2}\Big]\Big)\Big\}\Big].$$

# 4 Confidence Bands

## 4.1 Pointwise intervals vs. bands

For a fixed $t_0$, a $100(1-\alpha)\%$ CI $[L_{t_0}, U_{t_0}]$ satisfies $P\{L_{t_0} \le S(t_0) \le U_{t_0}\} = 1-\alpha$; once $t_0$ is fixed, $S(t_0)$ is just a scalar. Hence:

- a plot of pointwise intervals over all $t$ should *not* be read as an interval for the entire survival curve;
- such a plot is always "too narrow" compared with a band for the whole $S(t)$.

**Definition 6.7: Confidence band**

$[L(t), U(t)]$ is a $100(1-\alpha)\%$ confidence band (CB) for $S(t)$ on $t_L \le t \le t_U$ if

$$P\{L(t) \le S(t) \le U(t) \text{ for all } t_L \le t \le t_U\} = 1-\alpha,$$

where $t_L \ge$ the smallest and $t_U \le$ the largest observed event time.

## 4.2 Equal probability (EP) bands

- Define
$$a_L = \frac{n\sigma_S^2(t_L)}{1+n\sigma_S^2(t_L)}, \qquad a_U = \frac{n\sigma_S^2(t_U)}{1+n\sigma_S^2(t_U)}, \qquad 0 < a_L \le a_U < 1.$$
- EP bands are **proportional to the pointwise confidence bounds**.
- From Table C.3 (K&M Appendix C) find the coefficient $c_\alpha(a_L, a_U)$ and build the band over $[t_L, t_U]$ by replacing $Z_{1-\alpha/2}$ in Definition 6.1 and Theorems 6.2–6.3 with $c_\alpha(a_L, a_U)$. (E.g. for a 90% band, $c_{0.10}(0.10, 0.40) = 2.4721$, larger than $Z_{0.95} = 1.645$.)

## 4.3 Hall–Wellner (HW) bands

- **Not** proportional to the pointwise bounds: wider for small $t$ and narrower for large $t$.
- Find $k_\alpha(a_L, a_U)$ in Table C.4 and replace $Z_{1-\alpha/2}\sigma_S(t)$ in Definition 6.1 and Theorems 6.2–6.3 with
$$k_\alpha(a_L, a_U)\left[1+n\sigma_S^2(t)\right] n^{-1/2}.$$

## 4.4 Performance and sample-size advice

- The EP *linear* band performs poorly when $n < 200$: coverage is considerably below nominal.
- Log-log and arcsine-square-root EP bands have approximately correct coverage for $n \ge 20$; the arcsine EP band seems slightly better than the log-log EP band.
- For HW bands, all three versions perform reasonably well with as few as 20 observed events.

| EP | Sample size | HW | Sample size |
|---|---|---|---|
| Linear | $\ge 200$ | Linear | $\ge 20$ |
| Log-log | $\ge 20$ | Log-log | $\ge 20$ |
| Arcsine | $\ge 20$ | Arcsine | $\ge 20$ |

**Example 6.8: Bone marrow transplant, ALL patients (K&M Table 4.3)**

For 38 ALL patients, the KM estimate drops from $\hat{S}(1) = 0.9737$ ($Y = 38$) to $\hat{S}(662) = 0.3531$ ($Y = 13$), staying flat thereafter to the last (censored) time 2081; the Nelson–Aalen estimate reaches $\tilde{H}(662) = 1.0152$ with $\sigma_H = 0.2185$. Comparing 95% pointwise CIs, the EP band and the HW band for disease-free survival on roughly 100–600 days (K&M Fig. 4.5): both bands are wider than the pointwise intervals, and the HW band is wider than the EP band at early times but narrower later, as described above.

# Lecture 7. Estimation of Basic Quantities for Other Sampling Schemes

## Contents


## 1 Estimating the Mean Survival Time

### 1.1 Mean as the area under the curve

### 1.2 First solution: Efron's completion

### 1.3 Second solution: restricted mean


Since $\mu = \mathrm{E}(X) = \int_0^\infty S(t)\,\mathrm{d}t$, the natural estimator is $\hat{\mu} = \int_0^\infty \hat{S}(t)\,\mathrm{d}t$. Problem: if the last observation is censored, $\hat{S}(t)$ stays flat (never reaches 0) and the integral diverges. The right tail can be completed, but the estimate of the mean then depends on which completion is used.

Setting $\hat{S}(t) = 0$ for $t > t_{\max}$ (Efron) gives $\hat{\mu} = \int_0^{t_{\max}} \hat{S}(t)\,\mathrm{d}t$. Downsides: it assumes all survivors at $t_{\max}$ die immediately afterwards (biased downward), and $t_{\max}$ varies from sample to sample.

**Theorem 7.1: Restricted mean**

For a pre-specified $\tau$, estimate $\mu_\tau = \mathrm{E}[\min(X, \tau)] = \int_0^\tau S(t)\,\mathrm{d}t$ by

$$\hat{\mu}_\tau = \int_0^\tau \hat{S}(t)\,\mathrm{d}t, \qquad \hat{V}(\hat{\mu}_\tau) = \sum_{j=1}^{D} \Big[\int_{t_j}^\tau \hat{S}(t)\,\mathrm{d}t\Big]^2 \frac{d_j}{Y_j(Y_j - d_j)},$$

with $100(1-\alpha)\%$ CI $\hat{\mu}_\tau \pm Z_{1-\alpha/2}\sqrt{\hat{V}(\hat{\mu}_\tau)}$. (The sum runs over death times $t_j \le \tau$.)

**Remark 7.2: Why the weights $\int_{t_j}^\tau \hat{S}$**

A perturbation of the hazard estimate at $t_j$ rescales $\hat{S}(t)$ for every $t \ge t_j$, hence changes the area to the right of $t_j$ only. By the delta method, the contribution of $\hat{h}_j$ to $\mathrm{Var}(\hat{\mu}_\tau)$ is (area right of $t_j)^2 \times$ Greenwood's term $d_j/[Y_j(Y_j - d_j)]$.

Since $\hat{S}$ is a step function, $\hat{\mu}_\tau$ is a sum of rectangles: $\hat{\mu}_\tau = \sum_k \hat{S}(s_k)(s_{k+1} - s_k)$ over consecutive jump points $s_k$ in $[0, \tau]$.

**Example 7.3: 6-MP group (data of Lecture 5) [Supplement]**

KM: $\hat{S} = 0.857, 0.807, 0.753, 0.690, 0.627, 0.538, 0.448$ at $t_j = 6, 7, 10, 13, 16, 22, 23$ ($Y_j = 21, 17, 15, 12, 11, 7, 6$). With $\tau = 35$ (the largest, censored, time):

$$\begin{aligned}\hat{\mu}_{35} &= 6(1) + 1(0.857) + 3(0.807) + 3(0.753) + 3(0.690) \\ &\quad + 6(0.627) + 1(0.538) + 12(0.448) \approx 23.29 \text{ weeks},\end{aligned}$$

$$\hat{V}(\hat{\mu}_{35}) \approx 7.99, \quad \mathrm{SE} \approx 2.83, \quad 95\%\ \mathrm{CI} \approx (17.75,\ 28.83).$$

## 2 Quantiles and Their Confidence Intervals

Recall $x_p = \inf\{t : S(t) \le 1-p\}$ ($p = 0.5$: median). The estimator is $\hat{x}_p = \inf\{t : \hat{S}(t) \le 1-p\}$; its variance depends on the density $f(x_p)$, which is hard to estimate. Instead, CIs are obtained by **inverting** the pointwise CIs for $S(t)$.

**Theorem 7.4: CIs for $x_p$**

A $100(1-\alpha)\%$ CI for $x_p$ is the set of all $t$ satisfying

$$\text{linear:} \quad -Z_{1-\alpha/2} \le \frac{\hat{S}(t)-(1-p)}{\hat{V}^{1/2}[\hat{S}(t)]} \le Z_{1-\alpha/2},$$

$$\text{log-log:} \quad \left|\frac{[\log\{-\log\hat{S}(t)\}-\log\{-\log(1-p)\}]\,\hat{S}(t)\log\hat{S}(t)}{\hat{V}^{1/2}[\hat{S}(t)]}\right| \le Z_{1-\alpha/2},$$

$$\text{arcsine:} \quad \left|\frac{2\{\arcsin\sqrt{\hat{S}(t)}-\arcsin\sqrt{1-p}\}\{\hat{S}(t)(1-\hat{S}(t))\}^{1/2}}{\hat{V}^{1/2}[\hat{S}(t)]}\right| \le Z_{1-\alpha/2}.$$

**Derivation: where these come from**

Each is a Wald test of $H_0 : S(t) = 1-p$ on the transformed scale: $|g(\hat{S}) - g(1-p)|/(|g'(\hat{S})|\,\mathrm{SE}(\hat{S})) \le Z$. For log-log, $1/|g'(\hat{S})| = |\hat{S}\log\hat{S}|$; for arcsine-square-root, $1/|g'(\hat{S})| = 2\sqrt{\hat{S}(1-\hat{S})}$. The CI collects all $t$ at which $H_0$ is not rejected.

**Example 7.5: Median for the 6-MP group**

$\hat{S}(22) = 0.538 > 0.5 \ge \hat{S}(23) = 0.448$, so $\hat{x}_{0.5} = 23$. Checking the criteria at each death time, all three methods are satisfied from $t = 13$ onwards ($\hat{S}(13) = 0.690$, SE 0.107) but not at $t \le 10$. Since $\hat{S}$ never drops far enough below 0.5 to violate the criterion again, the 95% CI for the median is $[13, \infty)$: the upper limit cannot be determined.

## 3 Right-Censored and Left-Truncated Data

**Definition 7.6: Data and modified risk set**

Data $\{(L_i, T_i, \delta_i)\}$: $L_i$ = age at entering the study, $T_i$ = death or censoring time. At death times $t_1 < \cdots < t_D$ let $d_j$ = number of deaths and

$$Y_j = \sum_{i=1}^{n} I(L_i \le t_j \le T_i),$$

the number who have *entered* and are still under observation at $t_j$.

With this modified $Y_j$, all estimators of K&M Sections 4.2–4.4 (KM, Nelson–Aalen, Greenwood, CIs, bands) apply unchanged.

**Remark 7.7: What is being estimated**

Subjects are only seen after their entry time, so the estimators target *conditional* quantities:

$$S_L(t) = P(X > t \mid X > L), \qquad H_L(t) = -\ln P(X > t) + \ln P(X > L), \qquad H_L'(t) = h(t).$$

The hazard itself is unaffected by conditioning, which is why $d_j/Y_j$ still estimates it.

**Example 7.8: Channing House**

Residents of the Channing House retirement center (California): 97 males and 365 females; truncation times are the ages (months) at entry. The number at risk is small at young ages, rises to a peak around 950–1000 months, then falls.

- Males: the first subject entered at 751 and died at 777; the second entered at 759 and died at 781; the third appeared only at 782.
- Plain KM conditional on the youngest entry: at 777, $Y = 2$, $d = 1 \Rightarrow$ factor $1/2$; at 781, $Y = 1$, $d = 1 \Rightarrow$ factor 0. So $\hat{S}_L(t) = 0$ for all $t \geq 781$ — the estimate is destroyed by the tiny early risk set.
- Remedy: estimate $S_a(t) = P(X > t \mid X \geq a)$ for a chosen age $a$ where the risk set is reasonably large:

$$\hat{S}_a(t) = \prod_{a \leq t_j \leq t} \Big(1 - \frac{d_j}{Y_j}\Big), \quad \hat{V}[\hat{S}_a(t)] = \hat{S}_a(t)^2 \sum_{a \leq t_j \leq t} \frac{d_j}{Y_j(Y_j - d_j)},$$

$$\hat{H}_a(t) = \sum_{a \leq t_j \leq t} \frac{d_j}{Y_j}, \qquad t \geq a.$$

## 4 Left-Censored Data: Reversing the Time Scale

$X$ = failure time, $C$ = left censoring time; observe $T = \max(X, C)$, $\delta = I(T = X) = I(X \geq C)$. How to estimate $S(u) = P(X > u)$?

**Theorem 7.9: Reversed time-scale trick**

Pick $\tau > \max_i T_i$. Then $\tau - X_i$ is *right* censored by $\tau - C_i$:

$$\tau - T_i = \min(\tau - X_i,\ \tau - C_i), \qquad \delta_i = I(\tau - X_i \leq \tau - C_i).$$

Let $G(u) = P(\tau - X > u)$. Apply the usual KM to $\{(\tau - T_i, \delta_i)\}$ to get $\hat{G}$, and set

$$\hat{S}(u) = 1 - \hat{G}(\tau - u), \qquad \mathrm{Var}[\hat{S}(u)] = \mathrm{Var}[\hat{G}(\tau - u)].$$

CIs and bands are built the same way.

**Why [Supplement].** $\max(X, C) = \tau - \min(\tau - X, \tau - C)$, and $X \geq C \iff \tau - X \leq \tau - C$. Also $G(\tau - u) = P(\tau - X > \tau - u) = P(X < u)$, so $S(u) = 1 - G(\tau - u)$ (for continuous $X$). ■

# 5 Right-Truncated Data

We observe $\{(R_i, X_i)\}$ sampled conditionally on $X_i < R_i$. Reverse the time scale with $\tau > \max_i R_i$: then $\tau - X_i$ is *left truncated* by $\tau - R_i$ (since $X_i < R_i \iff \tau - X_i > \tau - R_i$). Use the standard left-truncation method to estimate the conditional distribution of $\tau - X_i$ given $\tau - X_i > a$.

**Definition 7.10: Choice $a = \tau - \tau^*$, $\tau^* = \max_i R_i$**

$$G^*(t) = P(\tau - X_i \geq t \mid \tau - X_i \geq \tau - \tau^*) = P(\tau - X_i \geq t \mid X_i \leq \tau^*),$$
$$S^*(t) = P(X_i > t \mid X_i \leq \tau^*) = 1 - G^*(\tau - t).$$

**Theorem 7.11: KM on the reversed scale**

With $\tau - t_j$ the $D$ distinct failure times on the reversed scale,

$$\hat{G}^*(t) = \prod_{\tau - \tau^* \leq \tau - t_j \leq t} \left(1 - \frac{d_j}{Y_j}\right),$$

$$d_j = \#\{i : \tau - X_i = \tau - t_j\}, \quad Y_j = \#\{i : \tau - X_i \geq \tau - t_j \geq \tau - R_i\},$$

and $\hat{S}^*(t) = 1 - \hat{G}^*(\tau - t)$. Inference for $S^*$ uses the properties of $\hat{G}^*$ for LT data.

Only the conditional survival given $X \leq \tau^*$ is estimable: nothing in the data speaks about $X > \tau^*$.

**Remark 7.12: Quick review**

KM and NA were developed for a single, possibly right-censored failure time with independent censoring. They extend directly to LT + RC data (modified risk set), and to left-censored or right-truncated data by reversing the time scale.

# 6 Other Sampling Schemes

Interval censoring, double censoring, and their combinations with left truncation have no simple closed-form estimator; Turnbull's iterative methods for these cases are covered in Lecture 8.

# Lecture 8. Estimation of Basic Quantities for Other Sampling Schemes (Part 2)

## Contents

## 1 Other Sampling Schemes

- Interval censoring: $X_i \in (L_i, R_i]$, never observed exactly.
- Double censoring: each observation is LC, RC, or exact.
- LT + interval censoring, or LT + double censoring.

Estimation here is hard: there is no simple closed-form estimator, and inference is not easy.

## 2 Turnbull's Iterative Method (1974) for Double Censoring

Pure left censoring is rare; samples with both left and right censoring are more common. Turnbull (1974) proposed a modified product-limit estimator with no closed form, computed iteratively.

- **Idea.** Start with an estimate of $S$ that ignores the left-censored data; use it to compute the *expected* number of deaths at each observed time contributed by the left-censored observations; update $S$; repeat until it stabilizes (a self-consistency / EM-type algorithm).
- **Notation.** $0 = t_0 < t_1 < \cdots < t_m$ the distinct observed times; $d_j$ = deaths at $t_j$ (possibly 0); $c_j$ = left-censored observations at $t_j$; $r_j$ = right-censored observations at $t_j$.

**Definition 8.1: Turnbull (1974) algorithm**

**Step 0.** Initial estimate $S_0(t_j)$, e.g. the KM ignoring the left-censored observations.

**Step** $(K{+}1).1$**.** For each left-censoring time $t_i$ and $j \le i$, estimate $p_{ij} = P(t_{j-1} < X \le t_j \mid X \le t_i)$ by

$$p_{ij} = \frac{S_K(t_{j-1}) - S_K(t_j)}{1 - S_K(t_i)}.$$

**Step** $(K{+}1).2$**.** Estimated events at $t_j$: $\hat{d}_j = d_j + \sum_{i=j}^{m} c_i\, p_{ij}$ (each left-censored observation is distributed over the times before it).

**Step** $(K{+}1).3$**.** Compute the usual KM from the pseudo right-censored data with $\hat{d}_j$ events and $r_j$ right-censored observations at $t_j$, ignoring the left-censored data: $Y_j = \sum_{l=j}^{m} (\hat{d}_l + r_l)$.

**Repeat** until $S_{K+1}(t_j) \approx S_K(t_j)$ for all $t_j$.

**Remark 8.2: Why $p_{ij}$ has this form**

A boy left censored at $t_i$ is known to have $X \le t_i$. Given that, the chance his event fell in $(t_{j-1}, t_j]$ is $P(t_{j-1} < X \le t_j)/P(X \le t_i) = [S(t_{j-1}) - S(t_j)]/[1 - S(t_i)]$. Replacing each left-censored observation by these fractional events is the E-step; recomputing KM is the

M-step.

**Example 8.3: Age at first marijuana use (K&M Section 1.17)**

California high-school boys: left censored = used marijuana but cannot recall the age of first use; right censored = never used. Initial KM ignoring the 12 left-censored boys (K&M Table 5.1):

| $t_i$ | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | $> 18$ | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $c_i$ | 0 | 0 | 0 | 1 | 2 | 3 | 2 | 3 | 1 | 0 | 12 |
| $d_i$ | 4 | 12 | 19 | 24 | 20 | 13 | 3 | 1 | 0 | 4 | 100 |
| $r_i$ | 0 | 0 | 2 | 15 | 24 | 18 | 14 | 6 | 0 | 0 | 79 |
| $Y_i$ | 179 | 175 | 163 | 142 | 103 | 59 | 28 | 11 | 4 | 4 | |
| $S_0(t_i)$ | .978 | .911 | .804 | .669 | .539 | .420 | .375 | .341 | .341 | 0 | |

Step 1 at $t_4 = 13$ (one left-censored boy, $1 - S_0(13) = 0.331$):

$$p_{41} = \tfrac{1.000-0.978}{0.331} = 0.067, \quad p_{42} = \tfrac{0.978-0.911}{0.331} = 0.202,$$

$$p_{43} = \tfrac{0.911-0.804}{0.331} = 0.320, \quad p_{44} = \tfrac{0.804-0.669}{0.331} = 0.410.$$

Doing this for every $i$ with $c_i > 0$ (K&M Table 5.2; e.g. the column $i = 9$ reads $0.034, 0.102, 0.161, 0.206, 0.197, 0.180, 0.068, 0.052, 0.000$), Step 2 gives

$$\hat{d}_1 = 4 + 0.067(1) + 0.048(2) + 0.039(3) + 0.036(2) + 0.034(3) + 0.034(1) = 4.487.$$

**Step 3 (K&M Table 5.3).**

| $t_i$ | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | $> 18$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $\hat{d}_i$ | 4.487 | 13.461 | 21.313 | 26.963 | 22.437 | 14.714 | 3.417 | 1.207 | 0 | 4 |
| $Y_i$ | 191.000 | 186.513 | 173.052 | 149.739 | 107.775 | 61.338 | 28.624 | 11.207 | 4 | 4 |
| $S_1(t_i)$ | .977 | .906 | .794 | .651 | .516 | .392 | .345 | .308 | .308 | 0 |

Iterate until convergence. Note $Y_1 = 191 = 179 + 12$: all left-censored boys are now accounted for as (fractional) events.

## 3 Turnbull's Iterative Method (1976) for Interval-Censored Data

Let $0 = \tau_0 < \tau_1 < \cdots < \tau_m$ be a grid containing all $L_i, R_i$. Define $\alpha_{ij} = 1$ if $(\tau_{j-1}, \tau_j] \subseteq (L_i, R_i]$ and 0 otherwise, i.e. whether the event of subject $i$ could have occurred at $\tau_j$.

**Definition 8.4: Turnbull (1976) algorithm**

**Step 1.** Probabilities of an event at $\tau_j$: $p_j = S(\tau_{j-1}) - S(\tau_j)$, $j = 1, \ldots, m$.

**Step 2.** Expected number of events at $\tau_j$: $d_j = \sum_{i=1}^{n} \frac{\alpha_{ij} p_j}{\sum_k \alpha_{ik} p_k}$ (each subject's unit mass is split over its admissible grid points in proportion to $p_j$).

**Step 3.** Estimated number at risk: $Y_j = \sum_{k=j}^{m} d_k$.

**Step 4.** Update the product-limit estimator from these pseudo-data; stop if it is close to the

previous $S$ at all $\tau_j$, else repeat Steps 1–3.

**Example 8.5: Time to cosmetic deterioration, early breast cancer (radiotherapy only)**

46 patients; data are intervals $(L, R]$ in months or right-censored "$\geq$" values: $(0, 7]$; $(0, 8]$; $(0, 5]$; $(4, 11]$; $(5, 12]$; $(5, 11]$; $(6, 10]$; $(7, 16]$; $(7, 14]$; $(11, 15]$; $(11, 18]$; $\geq 15$; $\geq 17$; $(17, 25]$; $(17, 25]$; $\geq 18$; $(19, 35]$; $(18, 26]$; $\geq 22$; $\geq 24$; $\geq 24$; $(25, 37]$; $(26, 40]$; $(27, 34]$; $\geq 32$; $\geq 33$; $\geq 34$; $(36, 44]$; $(36, 48]$; $\geq 36$; $\geq 36$; $(37, 44]$; $\geq 37$; $\geq 37$; $\geq 37$; $\geq 38$; $\geq 40$; $\geq 45$; $\geq 46$ ($\times 7$).

- **Initial estimate**: distribute each subject's mass $1/46$ equally over the grid points in $(L_i, R_i]$; e.g. the subject with $(0, 7]$ gives $(1/46)(1/4)$ to each of $\tau = 4, 5, 6, 7$.
- With $S_0$: $p_1 = 1 - 0.979 = 0.021$, $p_2 = 0.979 - 0.955 = 0.024$, etc.
- At $\tau = 4$, three intervals contain it ($(0, 8]$, $(0, 7]$, $(0, 5]$), so

$$d_1 = \frac{0.021}{0.021 + 0.024 + 0.021 + 0.029 + 0.031} + \frac{0.021}{0.021 + 0.024 + 0.021 + 0.029} + \frac{0.021}{0.021 + 0.024} = 0.842.$$

- One iteration changes $S$ only slightly at early times (e.g. $0.979 \to 0.982$ at $\tau = 4$) but more in the tail (e.g. $0.229 \to 0.341$ at $\tau = 46$); iterate to convergence.

# 4 The Cohort Life Table

## 4.1 Setting

A **cohort** is a group of individuals with a common origin from which event times are measured. Each event or censoring time is recorded only as falling in one of $k + 1$ adjacent, non-overlapping intervals $(a_{j-1}, a_j]$, $j = 1, \ldots, k + 1$. A traditional cohort life table presents the actual mortality experience of the cohort from the birth of each individual to the death of the last surviving member; it summarizes time-to-event information over these intervals (grouped data).

**Example 8.6: NLSY weaning data**

National Labor Survey of Youth (K&M Section 1.14): 927 first-born infants who were breast fed and have complete information. Response: duration of breast feeding in weeks, with an indicator of whether the baby was weaned (event) or lost to follow-up (censored). First rows: weeks 0–2: 927 entering, 2 withdrawn; 2–3: 848 entering, 3 withdrawn; 3–5: 774 entering, 6 withdrawn; ...

## 4.2 Construction

**Definition 8.7: Life-table quantities**

1. Intervals $I_j = (a_{j-1}, a_j]$, $j = 1, \ldots, k + 1$, $a_0 = 0$, $a_{k+1} = \infty$; each event/censoring time falls in exactly one of them.
2. $Y'_j$: number entering $I_j$ without having experienced the event.

3. $W_j$: number lost to follow-up or withdrawn alive in $I_j$.
4. $Y_j = Y_j' - W_j/2$: estimated number at risk in $I_j$, assuming censoring times are uniform over the interval (each withdrawal is at risk for half the interval on average) — the *actuarial* adjustment.
5. $d_j$: number of events in $I_j$.
6. Survival at the start of $I_j$ ($\hat{S}(a_0) = 1$):

$$\hat{S}(a_j) = \hat{S}(a_{j-1})[1 - d_j/Y_j] = \prod_{i=1}^{j} [1 - d_i/Y_i]. \tag{8.1}$$

7. Density at the midpoint $a_{mj} = (a_{j-1} + a_j)/2$:

$$\hat{f}(a_{mj}) = \frac{\hat{S}(a_{j-1}) - \hat{S}(a_j)}{a_j - a_{j-1}}. \tag{8.2}$$

8. Hazard at the midpoint (with $\hat{S}(a_{mj})$ taken as the average of the endpoint values):

$$\hat{h}(a_{mj}) = \frac{\hat{f}(a_{mj})}{\hat{S}(a_{mj})} = \frac{2\hat{f}(a_{mj})}{\hat{S}(a_j) + \hat{S}(a_{j-1})} = \frac{d_j}{(a_j - a_{j-1})(Y_j - d_j/2)}. \tag{8.3}$$

**Derivation: deriving (8.3)**

Write $\hat{q}_j = d_j/Y_j$, $\hat{p}_j = 1 - \hat{q}_j$ and $\Delta_j = a_j - a_{j-1}$. Then $\hat{S}(a_j) = \hat{S}(a_{j-1})\hat{p}_j$ and $\hat{f}(a_{mj}) = \hat{S}(a_{j-1})\hat{q}_j/\Delta_j$, so

$$\hat{h}(a_{mj}) = \frac{2\hat{S}(a_{j-1})\hat{q}_j/\Delta_j}{\hat{S}(a_{j-1})(1 + \hat{p}_j)} = \frac{2\hat{q}_j}{\Delta_j(1 + \hat{p}_j)} = \frac{2d_j/Y_j}{\Delta_j(2 - d_j/Y_j)} = \frac{d_j}{\Delta_j(Y_j - d_j/2)}.$$

Interpretation: events per unit of person-time, where those who fail in $I_j$ are credited with half the interval.

## 4.3 Standard errors

**Theorem 8.8: Estimated standard deviations**

$$\widehat{\mathrm{SD}}[\hat{S}(a_{j-1})] = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1} \frac{d_i}{Y_i(Y_i - d_i)}} = \hat{S}(a_{j-1})\sqrt{\sum_{i=1}^{j-1} \frac{\hat{q}_i}{Y_i\hat{p}_i}}, \tag{8.4}$$

$$\widehat{\mathrm{SD}}[\hat{f}(a_{mj})] = \frac{\hat{S}(a_{j-1})\hat{q}_j}{a_j - a_{j-1}}\sqrt{\sum_{i=1}^{j-1} \frac{\hat{q}_i}{Y_i\hat{p}_i} + \frac{\hat{p}_j}{Y_j\hat{q}_j}}, \tag{8.5}$$

$$\widehat{\mathrm{SD}}[\hat{h}(a_{mj})] = \left\{\frac{1 - [\hat{h}(a_{mj})(a_j - a_{j-1})/2]^2}{Y_j\hat{q}_j}\right\}^{1/2} \hat{h}(a_{mj}). \tag{8.6}$$

(8.4) is Greenwood's formula; the SD of $\hat{S}(a_0) = 1$ is 0.

### 4.4 Other summary quantities

1. **Mean**: $\hat{\mu} = \int_0^\infty \hat{S}(t)\,\mathrm{d}t$. If the largest observation is censored, $\hat{S}$ stays flat and the tail must be completed in one of the three ways of Lecture 5 (Efron, Gill, Brown–Hollander–Kowar).

2. **Median** by linear interpolation: find $I_j$ with $\hat{S}(a_{j-1}) \geq 0.5 \geq \hat{S}(a_j)$, then

$$\hat{x}_{0.5} = a_{j-1} + \frac{[\hat{S}(a_{j-1}) - 0.5](a_j - a_{j-1})}{\hat{S}(a_{j-1}) - \hat{S}(a_j)} = a_{j-1} + \frac{\hat{S}(a_{j-1}) - 0.5}{\hat{f}(a_{mj})}. \tag{8.7}$$

3. **Mean residual life** at the start of $I_i$:

$$\widehat{\mathrm{mr}}(a_{i-1}) = \frac{\int_{a_{i-1}}^\infty \hat{S}(t)\,\mathrm{d}t}{\hat{S}(a_{i-1})} = \frac{\sum_{j=i}^k \hat{S}(a_{j-1})(a_j - a_{j-1})}{\hat{S}(a_{i-1})}.$$

4. **Median residual life**: $\mathrm{mdrl}(a_{i-1})$ solves $S(a_{i-1})/2 = S\big(a_{i-1} + \mathrm{mdrl}(a_{i-1})\big)$ — the time increment beyond which half of those alive at $a_{i-1}$ are expected to survive.

**Example 8.9: life table for weaning (K&M Table 5.6)**

| Weeks | $Y'_j$ | $W_j$ | $Y_j$ | $d_j$ | $\hat{S}(a_{j-1})$ | $\hat{f}(a_{mj})$ | $\hat{h}(a_{mj})$ | SD $\hat{S}$ | SD $\hat{f}$ | SD $\hat{h}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0–2 | 927 | 2 | 926 | 77 | 1.0000 | 0.0416 | 0.0434 | 0 | 0.0045 | 0.0049 |
| 2–3 | 848 | 3 | 846.5 | 71 | 0.9168 | 0.0769 | 0.0875 | 0.0091 | 0.0088 | 0.0104 |
| 3–5 | 774 | 6 | 771 | 119 | 0.8399 | 0.0648 | 0.0836 | 0.0121 | 0.0055 | 0.0076 |
| 5–7 | 649 | 9 | 644.5 | 75 | 0.7103 | 0.0413 | 0.0618 | 0.0149 | 0.0046 | 0.0071 |
| 7–11 | 565 | 7 | 561.5 | 109 | 0.6276 | 0.0305 | 0.0537 | 0.0160 | 0.0027 | 0.0051 |
| 11–17 | 449 | 5 | 446.5 | 148 | 0.5058 | 0.0279 | 0.0662 | 0.0166 | 0.0021 | 0.0053 |
| 17–25 | 296 | 3 | 294.5 | 107 | 0.3381 | 0.0154 | 0.0555 | 0.0158 | 0.0014 | 0.0052 |
| 25–37 | 186 | 0 | 186 | 74 | 0.2153 | 0.0071 | 0.0414 | 0.0138 | 0.0008 | 0.0047 |
| 37–53 | 112 | 0 | 112 | 85 | 0.1296 | 0.0061 | 0.0764 | 0.0114 | 0.0006 | 0.0066 |
| 53– | 27 | 0 | 27 | 27 | 0.0313 | — | — | 0.0059 | — | — |

**Check of row 1**: $Y_1 = 927 - 2/2 = 926$; $\hat{q}_1 = 77/926 = 0.0832$, so $\hat{S}(2) = 0.9168$; $\hat{f} = (1 - 0.9168)/2 = 0.0416$; $\hat{h} = 77/\{2(926 - 38.5)\} = 0.0434$.

**Median**: $\hat{S}(11) = 0.5058 \geq 0.5 \geq \hat{S}(17) = 0.3381$, so $\hat{x}_{0.5} = 11 + \dfrac{(0.5058 - 0.5)(17 - 11)}{0.5058 - 0.3381} \approx$ 11.2 weeks.

**Hazard** (K&M Fig. 5.3): high (about 0.08–0.09 per week) in the first 5 weeks, about 0.05–0.07 through week 25, lowest (about 0.04) in weeks 25–37, then rising again in weeks 37–53.

# Lecture 9. Hypothesis Testing (Part 1)

## Contents

## 1 Overview

### 1.1 One-sample problems

**Rationale.** A special pattern in a KM curve may suggest a parametric model; parametric models are useful (efficiency; handling more complex sampling schemes), but once adopted we must check whether the model fits the data adequately. Hypotheses:

$$H_0 : S(t) = S_0(t), \qquad H_0 : H(t) = H_0(t), \qquad H_0 : h(t) = h_0(t) \qquad \text{for all } t \le \tau,$$

with $S_0$, $H_0$, $h_0$ *known*. E.g. under $X \sim \text{Exp}(\lambda)$: $H_0(t) = \lambda t$, $h_0(t) = \lambda$, $S_0(t) = e^{-\lambda t}$.

### 1.2 Two- and $K$-sample problems

The study population may be heterogeneous, with groups or key factors relevant to survival. With two groups we compare

$$H_0 : S_1(t) = S_2(t) \iff H_1(t) = H_2(t) \iff h_1(t) = h_2(t), \qquad t \in [0, \tau],$$

equivalent hypotheses with *no* known form. The $K$-sample case is the obvious extension: $H_0 : S_1(t) = \cdots = S_K(t)$, $t \in [0, \tau]$.

### 1.3 How to test?

Without censoring: one-sample $t$ test or Kolmogorov–Smirnov; two-/$K$-sample $t$ tests and ANOVA (approximately normal data) or Wilcoxon rank-sum and Kruskal–Wallis (skewed data). The difficulty is censoring (and truncation); our focus is censoring.

## 2 One-Sample Test with Right-Censored Data

Data $\{(T_i, \delta_i)\}$, $T_i = \min(X_i, C_i)$, $\delta_i = I(X_i \le C_i)$, $X_i$ with hazard $h(t)$. Test $H_0 : h(t) = h_0(t)$, equivalently $H_0 : H(t) = H_0(t)$, for $t \in [0, \tau]$, where $\tau$ is typically the largest observed study time. Notation: $t_1 < \cdots < t_D$ the distinct death times, $d_j = \sum_i I(T_i = t_j, \delta_i = 1)$, $Y_j = Y(t_j) = \sum_i I(T_i \ge t_j)$.

**Definition 9.1: One-sample weighted statistic**

Let $W(t)$ be a weight function with $W(t) = 0$ whenever $Y(t) = 0$. Compare the Nelson–Aalen increments with the hypothesized hazard:

$$Z(\tau) = O(\tau) - E(\tau) = \sum_{j=1}^{D} W(t_j) \frac{d_j}{Y(t_j)} - \int_0^\tau W(s) h_0(s)\, \mathrm{d}s.$$

Under $H_0$, its variance is

$$V[Z(\tau)] = \int_0^\tau W^2(s) \frac{h_0(s)}{Y(s)}\, \mathrm{d}s,$$

and for large samples $Z(\tau)^2/V[Z(\tau)] \sim \chi^2_1$.

**Remark 9.2: Intuition**

$\tilde{H}(t) = \sum_{t_j \le t} d_j/Y_j$ estimates $H(t)$; the statistic is a weighted distance $\int W \, \mathrm{d}(\tilde{H} - H_0)$. Each increment $d_j/Y_j$ has (Poisson) variance $\approx h_0 \, \mathrm{d}s/Y$, which gives $V[Z]$.

- The $\chi^2$ statistic only serves two-sided alternatives.
- One-sided: for $H_A : h(t) > h_0(t)$ for some $t \le \tau$, reject for large positive $T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$; for $H_A : h(t) < h_0(t)$, reject for large negative $T_1$.

## 2.1 One-sample log-rank test

**Theorem 9.3: $W(t) = Y(t)$: observed vs. expected**

Allow left truncation: $T_i$ = time on study, $L_i$ = delayed entry time. With $\tau$ the largest time on study,

$$O(\tau) = \sum_{j=1}^{D} Y(t_j) \frac{d_j}{Y(t_j)} = \sum_{j=1}^{D} \sum_{i=1}^{n} I(T_i = t_j, \delta_i = 1) = \sum_{i=1}^{n} I(\delta_i = 1),$$

$$E(\tau) = \int_0^\tau Y(s) h_0(s) \, \mathrm{d}s = \sum_{i=1}^{n} \int_0^\tau I(L_i \le s \le T_i) h_0(s) \, \mathrm{d}s = \sum_{i=1}^{n} [H_0(T_i) - H_0(L_i)].$$

$O$ = observed number of events at or before $\tau$; $E$ = expected number under $H_0$. Also $V[Z(\tau)] = \int_0^\tau Y(s) h_0(s) \, \mathrm{d}s = E(\tau)$, so the test statistic is $(O-E)^2/E$.

## 2.2 Harrington–Fleming weights

$$W_{HF}(t) = Y(t) S_0(t)^p [1 - S_0(t)]^q, \qquad p, q \ge 0, \quad S_0(t) = \exp[-H_0(t)].$$

- $p \gg q$: more weight on **early** departures from $H_0$;
- $q \gg p$: more weight on **late** departures;
- $p = q$: more weight on departures in the **mid-range**;
- $p = q = 0$: the one-sample log-rank test.

**Example 9.4: IBCSG Trial 6**

International Breast Cancer Study Group: a multi-center trial of adjuvant chemotherapy after surgical removal of the tumour; 735 subjects. The NA cumulative hazard plotted against time is roughly a straight line up to 100 months, suggesting an exponential distribution; eyeballing the slope gives $\hat{\lambda} \approx 0.005$. Formal one-sample log-rank tests ($W = Y$):

| $H_0$ | $T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$ | $p$-value |
|---|---|---|
| $H_0(t) = 0.0045t$ | $-0.002$ | 0.998 |
| $H_0(t) = 0.004t$ | 1.8 | 0.07 |
| $H_0(t) = 0.005t$ | $-1.4$ | 0.16 |
| $H_0(t) = 0.006t$ | $-4.1$ | $< 0.01$ |

$H_0(t) = 0.0045t$ fits best. (Sign check: a larger hypothesized rate inflates $E$, making $O - E$ negative.)

# 3 $K$-Sample Tests

## 3.1 Hypotheses

$$H_0 : h_1(t) = h_2(t) = \cdots = h_K(t) \ \ \forall t \le \tau \qquad \text{vs.} \qquad H_A : \text{ at least one } h_k(t) \text{ differs for some } t \le \tau.$$

Equivalently in terms of $H_k$ or $S_k$. Here $\tau$ is the largest time at which all groups have at least one subject at risk, typically $\tau = \min_k t_{\max,k}$. The alternative is *global*: reject if at least one population differs from the others at some time.

## 3.2 Data and notation

Right-censored data $\{(T_{ik}, \delta_{ik}),\ k = 1, \ldots, K,\ i = 1, \ldots, n_k\}$. Pool all data and let $0 < t_1 < \cdots < t_D$ be the distinct death times in the pooled sample:

- $d_j$, $d_{jk}$: deaths at $t_j$ in the pooled sample / in group $k$;
- $Y_j$, $Y_{jk}$: number at risk at $t_j$ in the pooled sample / in group $k$.

## 3.3 Basic idea and the test statistic

Under $H_0$ the hazards are equal but unknown, so both the pooled and within-group NA estimators estimate the same $H$:

$$\tilde{H}(t) = \sum_{t_j \le t} \frac{d_j}{Y_j} \to \int_0^t h(u)\,\mathrm{d}u, \qquad \tilde{H}_k(t) = \sum_{t_j \le t} I(Y_{jk} > 0)\frac{d_{jk}}{Y_{jk}} \to \int_0^t h(u)\,\mathrm{d}u.$$

Build a test from a weighted difference between $\tilde{H}_k$ and $\tilde{H}$.

**Definition 9.5: Weighted $K$-sample statistic**

With $W_k(t) \ge 0$ a random weight, zero when $Y_j = 0$ or $Y_{jk} = 0$:

$$Z_k(\tau) = \sum_{j=1}^{D} W_k(t_j)\Big\{\frac{d_{jk}}{Y_{jk}} - \frac{d_j}{Y_j}\Big\}.$$

Typically $W_k(t_j) = Y_{jk}W(t_j)$ with $W(\cdot)$ common to all groups, giving

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\} \quad = \sum_j W(t_j)\,(\text{observed} - \text{expected deaths in group } k).$$

### 3.4 Distribution under $H_0$

**Theorem 9.6: Asymptotic normality (via martingale theory)**

Under $H_0$, $(Z_1(\tau),\dots,Z_K(\tau)) \approx \mathrm{MVN}(\mathbf{0},\Sigma)$, $\Sigma = (\sigma_{kk'})_{K\times K}$, with

$$\hat\sigma_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1-\frac{Y_{jk}}{Y_j}\Big)\frac{Y_j-d_j}{Y_j-1}\,d_j, \qquad k=1,\dots,K,$$

$$\hat\sigma_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j-d_j}{Y_j-1}\,d_j, \qquad k\neq k'.$$

**Derivation: hypergeometric origin of the variance**

At $t_j$, condition on the margins: $Y_j$ at risk split into $(Y_{j1},\dots,Y_{jK})$ with $d_j$ deaths. Under $H_0$ the deaths are a random draw of $d_j$ out of $Y_j$ without replacement, so $(d_{j1},\dots,d_{jK})$ is multivariate hypergeometric with

$$\mathrm{E}(d_{jk}) = d_j\frac{Y_{jk}}{Y_j}, \quad \mathrm{Var}(d_{jk}) = d_j\frac{Y_{jk}}{Y_j}\Big(1-\frac{Y_{jk}}{Y_j}\Big)\frac{Y_j-d_j}{Y_j-1}, \quad \mathrm{Cov}(d_{jk},d_{jk'}) = -d_j\frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j-d_j}{Y_j-1}.$$

The factor $(Y_j-d_j)/(Y_j-1)$ is the finite-population correction (it equals 1 when $d_j=1$). Summing over (approximately uncorrelated) death times gives $\hat\sigma_{kk'}$.

### 3.5 Linear dependency and the final statistic

**Theorem 9.7: $\sum_k Z_k(\tau) = 0$**

$$\sum_{k=1}^{K} Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\sum_{k=1}^{K}\Big\{d_{jk} - d_j\frac{Y_{jk}}{Y_j}\Big\} = \sum_{j=1}^{D} W(t_j)(d_j-d_j) = 0,$$

since $\sum_k d_{jk} = d_j$ and $\sum_k Y_{jk} = Y_j$. Hence $Z_1,\dots,Z_K$ are linearly dependent and $\Sigma$ is singular.

Eliminate the redundancy by dropping one (arbitrary) $Z_K(\tau)$:

$$(Z_1,\dots,Z_{K-1}) \approx \mathrm{MVN}(\mathbf{0},\Sigma_{(K-1)\times(K-1)}),$$

and

$$\chi^2 = (Z_1(\tau),\dots,Z_{K-1}(\tau))\,\hat\Sigma^{-1}_{(K-1)\times(K-1)}\,(Z_1(\tau),\dots,Z_{K-1}(\tau))^T \overset{H_0}{\sim} \chi^2_{K-1}.$$

Reject at level $\alpha$ if $\chi^2 > \chi^2(K-1,1-\alpha)$. (The value does not depend on which $Z_k$ is dropped.)

### 3.6 Special case $K=2$

$$Z = \frac{Z_1(\tau)}{\sqrt{\hat V(Z_1(\tau))}} = \frac{\sum_{j=1}^{D} W(t_j)(d_{j1}-Y_{j1}d_j/Y_j)}{\sqrt{\sum_{j=1}^{D} W^2(t_j)\frac{Y_{j1}}{Y_j}(1-\frac{Y_{j1}}{Y_j})\frac{Y_j-d_j}{Y_j-1}d_j}} \overset{H_0}{\approx} N(0,1).$$

For $K=2$, a two-sided level-$\alpha$ test rejects if $|Z| > Z_{1-\alpha/2}$; one-sided tests are possible. For $K>2$ we must use the $\chi^2$ test, which only handles two-sided alternatives.

# 4 Choice of the Weight Function

Different weights change the power against specific alternatives; they are extremely well studied. All have the form $W_k(t_j) = Y_{jk}W(t_j)$.

1. **Log-rank**: $W(t) = 1$. Optimal power against **proportional hazards** alternatives $h_1(t) = c_2h_2(t) = \cdots = c_Kh_K(t)$.
2. **Gehan** (generalized Wilcoxon): $W(t) = Y(t) = \sum_k \sum_i I(T_{ik} \geq t)$, so $Z_k(\tau) = \sum_j (Y_j d_{jk} - d_j Y_{jk})$. Commonly used.
3. **Tarone–Ware**: $W(t) = f(Y(t))$ with known increasing $f$; $f(u) = u$ is Gehan, the original Tarone–Ware uses $f(u) = u^{1/2}$, also $f(u) = \log u$. Gives relatively larger weight to death times with larger risk sets.
4. **Peto–Peto**: $W(t) = \tilde{S}(t)$, $\tilde{S}(t) = \prod_{t_j \leq t} (1 - \frac{d_j}{Y_j+1})$, a KM-like estimator from the pooled sample.
5. **Modified Peto–Peto** (Andersen, 1982): $W(t) = \tilde{S}(t)Y(t)/(Y(t)+1)$. Both Peto–Peto tests depend on the pooled survival function.
6. **Fleming–Harrington**: $W(t_j) = \hat{S}(t_{j-1})^p [1 - \hat{S}(t_{j-1})]^q$, $p, q \geq 0$, $\hat{S}$ the pooled KM.
   - $p = q = 0$: log-rank;
   - $p > 0, q = 0$: larger weights early $\Rightarrow$ sensitive to early differences among hazards;
   - $p = 0, q > 0$: larger weights late $\Rightarrow$ sensitive to late differences.

**Remark 9.8: Why the left-continuous $\hat{S}(t_{j-1})$**

The weight at $t_j$ must be *predictable* — known just before $t_j$ — so it uses survival information up to $t_{j-1}$, not the deaths at $t_j$ themselves. This is what makes $Z_k$ a martingale-based statistic.

# 5 Catheter Placement in Kidney Dialysis Patients

Clinical trial (K&M Section 1.4) comparing time to cutaneous exit-site infection for catheters placed surgically (group 1) vs. percutaneously (group 2). From the KM curves (K&M Fig. 7.1), the surgical group does better early, but the percutaneous group has higher infection-free probability later (about 0.8 vs. 0.2 near month 27). The curves *cross*, so the choice of weight matters:

- weighting early times heavily (e.g. Gehan) may suggest the surgical group is better;
- weighting late times heavily may suggest the percutaneous group is better.

Relative weights (K&M Fig. 7.2): Gehan and Tarone–Ware decrease steeply with time; log-rank is flat; Fleming–Harrington with $q > 0$ increase with time.

**Example 9.9: Log-rank computation (K&M Table 7.2, selected rows)**

| $t_j$ | $Y_{j1}$ | $d_{j1}$ | $Y_{j2}$ | $d_{j2}$ | $Y_j$ | $d_j$ | $Y_{j1}\frac{d_j}{Y_j}$ | $d_{j1} - Y_{j1}\frac{d_j}{Y_j}$ | $\frac{Y_{j1}}{Y_j}\left(1-\frac{Y_{j1}}{Y_j}\right)\frac{Y_j-d_j}{Y_j-1}d_j$ |
|---|---|---|---|---|---|---|---|---|---|
| 0.5 | 43 | 0 | 76 | 6 | 119 | 6 | 2.168 | −2.168 | 1.326 |
| 1.5 | 43 | 1 | 60 | 0 | 103 | 1 | 0.417 | 0.583 | 0.243 |
| 2.5 | 42 | 0 | 56 | 2 | 98 | 2 | 0.857 | −0.857 | 0.485 |
| 4.5 | 36 | 2 | 43 | 0 | 79 | 2 | 0.911 | 1.089 | 0.490 |
| | | | | | | | ⋮ | | |
| 26.5 | 2 | 1 | 3 | 0 | 5 | 1 | 0.400 | 0.600 | 0.240 |
| SUM | | 15 | | 11 | | 26 | 11.036 | 3.964 | 6.211 |

So $Z_1(\tau) = 15 - 11.036 = 3.964$, $\hat{\sigma}_{11} = 6.211$, $\chi^2 = 3.964^2/6.211 = 2.53$, $p = 0.112$: no significant difference by the log-rank test.

**Example 9.10: Comparison of two-sample tests (K&M Table 7.3)**

| Test | $W(t_j)$ | $Z_1(\tau)$ | $\sigma^2_{11}$ | $\chi^2$ | $p$-value |
|---|---|---|---|---|---|
| Log-rank | 1 | 3.96 | 6.21 | 2.53 | 0.112 |
| Gehan | $Y_j$ | −9 | 38862 | 0.002 | 0.964 |
| Tarone–Ware | $Y_j^{1/2}$ | 13.20 | 432.83 | 0.40 | 0.526 |
| Peto–Peto | $\tilde{S}(t_j)$ | 2.47 | 4.36 | 1.40 | 0.237 |
| Modified Peto–Peto | $\tilde{S}(t_j)Y_j/(Y_j+1)$ | 2.31 | 4.20 | 1.28 | 0.259 |
| FH $p=0, q=1$ | $1-\hat{S}(t_{j-1})$ | 1.41 | 0.21 | 9.67 | 0.002 |
| FH $p=1, q=0$ | $\hat{S}(t_{j-1})$ | 2.55 | 4.69 | 1.39 | 0.239 |
| FH $p=1, q=1$ | $\hat{S}(t_{j-1})[1-\hat{S}(t_{j-1})]$ | 1.02 | 0.11 | 9.83 | 0.002 |
| FH $p=0.5, q=0.5$ | $\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^{0.5}$ | 2.47 | 0.66 | 9.28 | 0.002 |
| FH $p=0.5, q=2$ | $\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^2$ | 0.32 | 0.01 | 8.18 | 0.004 |

**Remark 9.11: Comments**

- Gehan's weight puts very heavy weight on early comparisons, leading to a *negative* $Z_1(\tau)$.
- Fleming–Harrington tests with $q > 0$ weight late comparisons and are significant, because the two curves diverge for large $t$.
- In practice, the choice of weight is an important decision and should be made *before* looking at the data.
- The log-rank test ($W = 1$) is the most popular, followed by Gehan (generalized Wilcoxon rank-sum); both are in most statistical packages.

## 6 Three-Group Comparison (Bone Marrow Transplant)

Disease-free survival of leukaemia patients after bone marrow transplantation: Group 1 = 38 ALL, Group 2 = 54 AML low-risk, Group 3 = 45 AML high-risk. Test $H_0 : S_1(t) = S_2(t) = S_3(t)$, $t \le \tau = 2204$ days.

**Example 9.12: Log-rank test with $K = 3$**

Using the first two statistics, $Z_1(2204) = 2.148$, $Z_2(2204) = -14.966$, with

$$\hat{\Sigma} = \begin{pmatrix} 15.9552 & -10.3451 \\ -10.3451 & 20.3398 \end{pmatrix}, \qquad \chi^2 = (2.148,\ -14.966)\,\hat{\Sigma}^{-1} \begin{pmatrix} 2.148 \\ -14.966 \end{pmatrix} = 13.8037.$$

On $K - 1 = 2$ df, $p \approx 0.001$.

**Derivation: evaluating the quadratic form [Supplement]**

$\det \hat{\Sigma} = 15.9552(20.3398) - 10.3451^2 = 324.53 - 107.02 = 217.51$, and $\hat{\Sigma}^{-1} = \frac{1}{217.51}\begin{pmatrix} 20.3398 & 10.3451 \\ 10.3451 & 15.9552 \end{pmatrix}$. Then

$$\begin{aligned} \chi^2 &= \frac{1}{217.51}\Big[20.3398(2.148)^2 + 2(10.3451)(2.148)(-14.966) + 15.9552(-14.966)^2\Big] \\ &= \frac{93.85 - 665.14 + 3573.73}{217.51} \approx 13.80. \end{aligned}$$

Other weights: Gehan $\chi^2 = 16.24$ ($p = 0.0003$); Tarone–Ware 15.65 ($p = 0.0004$); FH $(1, 0)$ 15.67 ($p = 0.0004$); FH $(0, 1)$ 6.11 ($p = 0.047$); FH $(1, 1)$ 9.93 ($p = 0.007$). The results are robust: all tests suggest that the disease-free survival curves differ among the three groups.

# Lecture 10. Hypothesis Testing (Part 2)

## Contents

## 1 Review: The $K$-Sample Test

Right-censored data $\{(T_{ik}, \delta_{ik})\}$; pooled death times $t_1 < \cdots < t_D$; $d_j, d_{jk}$ deaths and $Y_j, Y_{jk}$ at risk at $t_j$ (pooled / group $k$). With a common weight $W$,

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\}, \qquad \chi^2 = (Z_1, \ldots, Z_{K-1})\,\hat{\Sigma}_{K-1}^{-1}\,(Z_1, \ldots, Z_{K-1})^T \overset{H_0}{\approx} \chi^2_{K-1},$$

with

$$\hat{\sigma}_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}d_j, \qquad \hat{\sigma}_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}d_j.$$

This omnibus test is **not sensitive to a trend alternative**.

## 2 Tests for Trend

### 2.1 Order-specific alternatives

$$H_0 : h_1(t) = \cdots = h_K(t) \;\; \forall t \le \tau \qquad \text{vs.} \qquad H_A : h_1(t) \le h_2(t) \le \cdots \le h_K(t) \;\; \forall t \le \tau,$$

with at least one strict inequality; equivalently $H_A : S_1(t) \ge S_2(t) \ge \cdots \ge S_K(t)$.

### 2.2 Trend test statistic

**Objective**: increase power by a statistic sensitive to the stochastic ordering.

**Definition 10.1: Trend test**

For scores $a_1 < a_2 < \cdots < a_K$,

$$Z = \frac{\sum_{k=1}^{K} a_k Z_k(\tau)}{\Big(\sum_{k=1}^{K}\sum_{k'=1}^{K} a_k a_{k'} \hat{\sigma}_{kk'}\Big)^{1/2}} = \frac{\mathbf{a}'\mathbf{Z}}{\sqrt{\mathbf{a}'\hat{\Sigma}\mathbf{a}}},$$

with $\hat{\sigma}_{kk'}$ from the full $K \times K$ $\hat{\Sigma}$. Under $H_0$ each $Z_k(\tau)$ is a mean-zero martingale, so the linear combination is approximately normal with mean 0, and $Z \approx N(0,1)$. Reject $H_0$ in favor of $H_A$ at level $\alpha$ if $Z$ exceeds the upper $\alpha$ quantile of $N(0,1)$ (one-sided).

- **Choice of** $a_k$: heuristically, under $H_A$, $\mathrm{E}\{Z_1(\tau)\} \le \mathrm{E}\{Z_2(\tau)\} \le \cdots \le \mathrm{E}\{Z_K(\tau)\}$ (higher hazard $\Rightarrow$ more observed than expected deaths). Taking $a_k = k$ (standard) gives large weight to the groups with large expected values under $H_A$. Other choices may have better properties.
- The full $\hat{\Sigma}$ is singular, but $\mathbf{a}'\hat{\Sigma}\mathbf{a} > 0$ as long as $\mathbf{a}$ is not constant, so no group needs to be dropped.

### 2.3 Invariance

**Theorem 10.2: Invariance under linear transformations of the scores**

Let $Z_{\mathbf{a}} = \mathbf{a}'\mathbf{Z}/\sqrt{V[\mathbf{a}'\mathbf{Z}]}$ and $\mathbf{b} = c\mathbf{a} + d\mathbf{1}_K$ with constants $c > 0$, $d$. Then $Z_{\mathbf{b}}(\tau) = Z_{\mathbf{a}}(\tau)$.

**Proof [Supplement].** Since $\sum_k Z_k(\tau) = 0$, i.e. $\mathbf{1}'\mathbf{Z} = 0$ (and hence $\hat{\Sigma}\mathbf{1} = \mathbf{0}$),

$$\mathbf{b}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z} + d\,\mathbf{1}'\mathbf{Z} = c\,\mathbf{a}'\mathbf{Z}, \qquad V[\mathbf{b}'\mathbf{Z}] = V[c\,\mathbf{a}'\mathbf{Z}] = c^2 V[\mathbf{a}'\mathbf{Z}].$$

So $Z_{\mathbf{b}} = c\,\mathbf{a}'\mathbf{Z}/(c\sqrt{V[\mathbf{a}'\mathbf{Z}]}) = Z_{\mathbf{a}}$. (If $c < 0$ the sign flips.) ■

E.g. scores $(1, 2, 3, 4)$ and $(0, 1, 2, 3)$ or $(-3, -1, 1, 3)$ give the same test.

**Example 10.3: larynx cancer by stage**

90 larynx cancer patients in stages I–IV. KM curves (K&M Fig. 7.3) are ordered: stage I best, stage IV worst. Give the most severe disease the heaviest weight: $a_k = k$, $k = 1, \ldots, 4$. With log-rank weights,

$$\mathbf{Z}(10.7) = (-7.5660,\ -3.0117,\ 2.9155,\ 7.6623),$$

$$\hat{\Sigma} = \begin{pmatrix} 12.0740 & -4.4516 & -6.2465 & -1.3759 \\ -4.4516 & 7.8730 & -2.7599 & -0.6614 \\ -6.2465 & -2.7599 & 9.9302 & -0.9238 \\ -1.3759 & -0.6614 & -0.9238 & 2.9612 \end{pmatrix}.$$

Then $\mathbf{a}'\mathbf{Z} = -7.566 - 6.023 + 8.747 + 30.649 = 25.806$ and $\mathbf{a}'\hat{\Sigma}\mathbf{a} = 48.152$, so

$$Z = \frac{25.806}{\sqrt{48.152}} = 3.72, \qquad Z^2 = 13.83, \qquad p < 0.0001.$$

(The value 13.83 reported in K&M is $Z^2$.) With other weights: Tarone–Ware $Z = 4.06$, Gehan 4.22, Peto–Peto 4.13. All have $p < 0.0001$, strong evidence that the hazard rates are in the expected order.

**Remark 10.4**

Use a trend test only with strong prior information about the group ordering; otherwise it can lose power relative to the omnibus test.

## 3 Stratified Tests

### 3.1 Rationale

When comparing $K$ groups we often need to adjust for other covariates. E.g. compare 3 vs. 6 months of chemotherapy in IBCSG adjusting for tumour stage (I–IV): the interest is the treatment, not the stage. One option is a regression model for covariate effects, but model-based adjustment is sensitive to misspecification. A conservative alternative is **stratification**.

### 3.2 Set-up

**Definition 10.5: Stratified $K$-sample test**

$H_0 : h_{1s}(t) = h_{2s}(t) = \cdots = h_{Ks}(t)$, $t \in [0, \tau]$, for each stratum $s = 1, \ldots, M$.

1. Using only the data in stratum $s$, compute $Z_{ks}(\tau)$, $k = 1, \ldots, K$, and $\hat{\Sigma}_s = \widehat{\mathrm{Cov}}[Z_{1s}(\tau), \ldots, Z_{Ks}(\tau)]$.
2. Global test: $Z_{k\cdot}(\tau) = \sum_{s=1}^{M} Z_{ks}(\tau)$ and $\hat{\Sigma}_{\cdot} = \widehat{\mathrm{Cov}}[Z_{1\cdot}, \ldots, Z_{K\cdot}] = \sum_{s=1}^{M} \hat{\Sigma}_s$ (strata are independent).
3. $\chi^2 = [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]\, \hat{\Sigma}^{-1}_{(K-1)\times(K-1)}\, [Z_{1\cdot}, \ldots, Z_{(K-1)\cdot}]^T \overset{H_0}{\sim} \chi^2_{K-1}$.

Within each stratum, observed and expected deaths are compared only among subjects with the same stratum, so differences in the baseline hazard across strata cannot masquerade as group effects.

**Example 10.6: bone marrow transplant, stratified by MTX**

The three disease groups (ALL, AML low risk, AML high risk) were compared in Lecture 9 and all tests rejected $H_0$. A more careful comparison accounts for whether patients received the graft-versus-host prophylactic methotrexate (MTX) or not (NOMTX). Stratified log-rank ($W = 1$):

$$\text{MTX:} \quad (Z_1, Z_2, Z_3) = (1.2816,\ -2.4929,\ 1.2113),$$
$$\hat{\Sigma}_{MTX} = \begin{pmatrix} 6.23336 & -3.42915 & -2.80422 \\ -3.42915 & 5.60732 & -2.17818 \\ -2.80422 & -2.17818 & 4.98239 \end{pmatrix},$$
$$\text{NOMTX:} \quad (Z_1, Z_2, Z_3) = (-0.504,\ -11.235,\ 11.739),$$
$$\hat{\Sigma}_{NOMTX} = \begin{pmatrix} 9.6771 & -6.5535 & -3.1236 \\ -6.5535 & 13.9816 & -7.4281 \\ -3.1236 & -7.4281 & 10.5517 \end{pmatrix}.$$

Summing: $Z_{1\cdot} = 0.7776$, $Z_{2\cdot} = -13.7279$, $Z_{3\cdot} = 12.9503$ (again linearly dependent), and

$$\hat{\Sigma}_{\cdot} = \begin{pmatrix} 15.91046 & -9.98265 & -5.92782 \\ -9.98265 & 19.58892 & -9.60628 \\ -5.92782 & -9.60628 & 15.53409 \end{pmatrix}.$$

Using the first two:

$$\chi^2 = (0.7776,\ -13.7279) \begin{pmatrix} 15.9105 & -9.9827 \\ -9.9827 & 19.5889 \end{pmatrix}^{-1} \begin{pmatrix} 0.7776 \\ -13.7279 \end{pmatrix} = 13.19 \quad (2 \text{ df},\ p \approx 0.001).$$

**Remark 10.7: Remarks**

Stratified tests have good power against alternatives in the *same direction* in every stratum. Otherwise, $Z$ from one stratum can be cancelled by $Z$ from another, the overall statistic may be near 0, and the test has very low power. In that case, run separate $K$-sample tests in each stratum.

# 4 Matched Pairs

Another use of stratification: each matched pair is a stratum. Paired times $(T_{1i}, T_{2i})$ and indicators $(\delta_{1i}, \delta_{2i})$, $i = 1, \dots, M$; test $H_0 : h_{1i}(t) = h_{2i}(t)$. With $K = 2$, in each pair

$$Z_{1i}(\tau) = \sum_j W(t_j)\Big\{d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big\}, \qquad \sigma_{11i} = \sum_j W(t_j)^2 \frac{Y_{j1}}{Y_j}\Big(1 - \frac{Y_{j1}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1} d_j.$$

**Derivation: within a single pair**

Before the first time in the pair both are at risk: $Y_j = 2$, $Y_{j1} = 1$, so $Y_{j1}/Y_j = 1/2$.

1. $T_{1i} = T_{2i}$: $d_j = \delta_{1i} + \delta_{2i}$,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i} + \delta_{2i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i} - \delta_{2i}}{2},$$

$$\sigma_{11i} = W^2(T_{1i})\tfrac{1}{2}(1 - \tfrac{1}{2})\frac{2 - \delta_{1i} - \delta_{2i}}{2 - 1}(\delta_{1i} + \delta_{2i})$$

$$= \frac{W^2(T_{1i})}{4}(2 - \delta_{1i} - \delta_{2i})(\delta_{1i} + \delta_{2i}).$$

2. $T_{1i} < T_{2i}$: only the first time can be an event with both at risk,

$$Z_{1i} = W(T_{1i})\Big\{\delta_{1i} - \frac{\delta_{1i}}{2}\Big\} = W(T_{1i})\frac{\delta_{1i}}{2}, \qquad \sigma_{11i} = \frac{W^2(T_{1i})}{4}(2 - \delta_{1i})\delta_{1i}.$$

3. $T_{1i} > T_{2i}$: $Z_{1i} = -W(T_{2i})\dfrac{\delta_{2i}}{2}$, $\quad \sigma_{11i} = \dfrac{W^2(T_{2i})}{4}(2 - \delta_{2i})\delta_{2i}$.

(After the first time, one of the pair is gone, so $Y_{j1}/Y_j \in \{0, 1\}$ and later terms contribute nothing.) For 0/1 indicators $(2 - \delta)\delta = \delta$, so:

$$Z_{1i} = \begin{cases} W(T_{1i})/2, & T_{1i} < T_{2i}, \delta_{1i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{1i} = 1, \delta_{2i} = 0, \\ -W(T_{2i})/2, & T_{2i} < T_{1i}, \delta_{2i} = 1 \text{ or } T_{1i} = T_{2i}, \delta_{2i} = 1, \delta_{1i} = 0, \\ 0, & \text{otherwise;} \end{cases}$$

$$\hat\sigma_{11i} = \begin{cases} W(T_{1i})^2/4, & \text{first case,} \\ W(T_{2i})^2/4, & \text{second case,} \\ 0, & \text{otherwise.} \end{cases}$$

**Theorem 10.8: Paired log-rank (sign-type) test**

If $W$ does not depend on time (e.g. log-rank, $W \equiv 1$), with $D_1 = \#$ pairs in which member 1 has the event first and $D_2 = \#$ pairs in which member 2 does,

$$Z_{1\cdot} = \frac{W}{2}(D_1 - D_2), \qquad \hat\sigma_{11\cdot} = \sum_{i=1}^{M}\hat\sigma_{11i} = \frac{W^2}{4}(D_1 + D_2), \qquad \frac{Z_{1\cdot}}{\sqrt{\hat\sigma_{11\cdot}}} = \frac{D_1 - D_2}{\sqrt{D_1 + D_2}} \overset{H_0}{\approx} N(0, 1).$$

This is the sign test (McNemar-type): under $H_0$ each informative pair is equally likely to go either way.

**Example 10.9: 6-MP vs. placebo matched pairs**

$D_{\text{Placebo}} = 18$, $D_{\text{6-MP}} = 3$, so $Z = (18-3)/\sqrt{18+3} = 3.27$ and $p = 2P[Z > 3.27] = 0.001$: sufficient evidence that the relapse rates differ.

# 5 Renyi-Type Tests

## 5.1 Rationale

If the hazards of the two groups **cross**, the usual two-sample tests may have low power: early differences cancel late differences of opposite sign. Renyi-type tests are built on the *absolute values of cumulative differences*; a Renyi statistic is the analogue for censored data of the Kolmogorov–Smirnov statistic.

## 5.2 Set-up and statistic

Test $H_0 : h_1(t) = h_2(t)$, $t \le \tau$, vs. $H_A : h_1(t) \neq h_2(t)$ for some $t \le \tau$. Two samples of sizes $n_1, n_2$; pooled distinct failure times $t_1 < \cdots < t_D$; $d_{j1}, d_{j2}$ and $Y_{j1}, Y_{j2}$ with $d_j = d_{j1} + d_{j2}$, $Y_j = Y_{j1} + Y_{j2}$.

**Definition 10.10: Renyi statistic**

Cumulative weighted difference up to $t_i$:

$$Z(t_i) = \sum_{t_j \le t_i} W(t_j)\Big[d_{j1} - Y_{j1}\frac{d_j}{Y_j}\Big], \qquad i = 1, \ldots, D,$$

with standard error at $\tau$ (largest $t_j$ with $Y_{j1}, Y_{j2} > 0$)

$$\sigma^2(\tau) = \sum_{t_j \le \tau} W^2(t_j)\frac{Y_{j1}}{Y_j}\frac{Y_{j2}}{Y_j}\frac{Y_j - d_j}{Y_j - 1}d_j.$$

The two-sided test statistic is

$$Q = \frac{\sup\{|Z(t)|,\ t \le \tau\}}{\sigma(\tau)}.$$

**Theorem 10.11: Null distribution**

Under $H_0$, $Q$ is approximately distributed as $\sup\{|B(x)|,\ 0 \le x \le 1\}$, $B$ a standard Brownian motion. Critical values are in K&M Table C.5.

**Remark 10.12: Why Brownian motion**

$Z(t)$ is a martingale in $t$ with variance process $\sigma^2(t)$; rescaling time by $x = \sigma^2(t)/\sigma^2(\tau) \in [0, 1]$, $Z/\sigma(\tau)$ behaves like $B(x)$. The supremum of $|Z|$ catches a large excursion even if $Z(\tau)$ returns near 0.

**Comparison with the usual test.** The usual statistic is $Z(\tau)/\sqrt{\text{Var}[Z(\tau)]}$ (or its square). When hazards cross, early positive differences are cancelled by later negative ones, and the log-rank test may lack power; the supremum test based on $|Z|$ has greater power for such differences.

**Example 10.13: gastric cancer: chemotherapy vs. chemotherapy + radiotherapy**

The KM curves cross (K&M Fig. 7.5): chemo + radiation is worse early but better later. Using the log-rank version of the Renyi statistic, $|Z(t_j)|$ (K&M Fig. 7.4) peaks at $t_j = 315$ with value 9.8; $\sigma(\tau = 2363) = 4.46$, so

$$Q = 9.8/4.46 = 2.2, \qquad 0.05 < p < 0.06 \text{ (K\&M Table C.5)},$$

close to rejecting $H_0$. The usual log-rank test gives $Z(2363) = -2.15$, statistic $-2.15/4.46 = -0.482$, $p = 2P(Z < -0.482) = 0.63$: it misses the crossing difference entirely.